MATH868C
Several Complex Variables



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1 Review

Definition 1.1 C^1 function $f : \Omega \to \mathbb{C}$ is *holomorphic* if $\bar{\partial} f = 0$. Denote the set of all holomorphic functions on Ω as $A(\Omega)$

Lemma 1.2 If f is holomorphic, then $\int_{\partial\Omega} f dz = 0$

Proof.

$$\int_{\partial\Omega} f dz = \int_{\Omega} d(f dz) = \int_{\Omega} \bar{\partial} f \wedge dz = 0$$

□

Theorem 1.3 (Poincaré-Lelong formula) Since $\Delta = \partial_x^2 + \partial_y^2 = 4\partial_z\partial_{\bar{z}} = 4\partial_{\bar{z}}\partial_z$, $dz \wedge d\bar{z} = -2idx \wedge dy = -2id\mu$. In the distributional sense, $-\frac{\log r}{2\pi} = -\frac{1}{4\pi}\log(x^2+y^2)$ is the fundamental solution of Laplacian equation in dimension 2, i.e. $\Delta \log(x^2+y^2) = 4\pi\delta$, we have

$$\Delta \log|\ z\ |^2\ dz \wedge d\bar{z} = 4\pi\delta dz \wedge d\bar{z} \Leftrightarrow \bar{\partial}\partial \log|\ z\ |^2 = 2\pi i \delta dx \wedge dy$$

Note. $\partial \log|\ z\ |^2 = \partial \log(z) + \partial \log(\bar{z}) = \frac{dz}{z}$ is integrable around 0

Proof. We prove a slightly general result. For any $\phi \in C_c^\infty(\Omega)$, by definition we have

$$\begin{aligned}
\iint_\Omega \phi\bar{\partial}\partial \log|\ z-w\ |^2 &= -\iint_\Omega \bar{\partial}\phi \wedge \partial \log|\ z-w\ |^2 \\
&= -\lim_{\epsilon\to 0} \iint_{|\ z-w\ |\ \geq\epsilon} \bar{\partial}\phi \wedge \partial \log|\ z-w\ |^2 \\
&= -\lim_{\epsilon\to 0} \iint_{|\ z-w\ |\ \geq\epsilon} d(\phi\partial \log|\ z-w\ |^2) \\
&= \lim_{\epsilon\to 0} \oint_{|\ z-w\ |\ =\epsilon} \phi\partial \log|\ z-w\ |^2 \\
&= \lim_{\epsilon\to 0} \oint_{|\ z-w\ |\ =\epsilon} \frac{\phi}{z-w} dz \\
&= 2\pi i\phi(w)
\end{aligned}$$

□

Theorem 1.4 (Cauchy's formula) If $f \in C^1(\overline{\Omega})$, then

$$f(w) = \frac{1}{2\pi i}\iint_\Omega \frac{\partial_{\bar{z}} f dz \wedge d\bar{z}}{z-w} + \frac{1}{2\pi i}\int_{\partial\Omega} \frac{f}{z-w} dz$$

Proof. By Poincaré-Lelong formula 1.3, we have

$$\begin{aligned}
f(w) &= \frac{1}{2\pi i}\iint_\Omega f\bar{\partial}\partial \log|\ z-w\ |^2 \\
&= -\frac{1}{2\pi i}\iint_\Omega \bar{\partial} f \wedge \partial \log|\ z-w\ |^2 + \frac{1}{2\pi i}\int_{\partial\Omega} f\partial \log|\ z-w\ |^2 \\
&= \frac{1}{2\pi i}\iint_\Omega \frac{\partial_{\bar{z}} f dz \wedge d\bar{z}}{z-w} + \frac{1}{2\pi i}\int_{\partial\Omega} \frac{f}{z-w} dz
\end{aligned}$$

□

Corollary 1.5 If $f \in C^1(\overline{\Omega}) \cap A(\Omega)$, then by Cauchy's formula 1.4, we know

$$f(w) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(z)}{z-w} dz$$

Which is C^∞ in w

$$f^{(n)}(w) = \frac{n!}{2\pi i} \int_{\partial\Omega} \frac{f(z)}{(z-w)^{n+1}} dz$$

Corollary 1.6 (Cauchy's estimate) For $K \subseteq \Omega$ compact, there are constants C_n such that for any $f \in A(\Omega)$

$$\sup_{z \in K} | \ f^{(n)}(z) \ | \leq C_n \parallel f \parallel_{L^1(\Omega)}$$

Proof. Consider a bump function χ with $\operatorname{supp} \chi \subseteq \Omega$ and $\chi \equiv 1$ on K, then for any $w \in K$

$$\begin{aligned} f(w) &= \frac{1}{2\pi i} \iint_{\Omega} \frac{\partial_{\bar{z}}(\chi f) dz \wedge d\bar{z}}{z-w} + \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\chi f}{z-w} dz \\ &= \frac{1}{2\pi i} \iint_{\Omega} \frac{(\partial_{\bar{z}}\chi) f dz \wedge d\bar{z}}{z-w} \\ &= \frac{1}{2\pi i} \iint_{\Omega \setminus K} \frac{(\partial_{\bar{z}}\chi) f dz \wedge d\bar{z}}{z-w} \end{aligned}$$

$\frac{\partial_{\bar{z}}\chi}{z-w}$ can be bounded on $\Omega \setminus K$

□

Corollary 1.7 $A(\Omega) \subseteq C(\Omega)$ is closed, thus a Fréchet space

Proof. Suppose $\{f_j\} \subseteq A(\Omega)$ converges to f in $C(\Omega)$, but since

$$f_j(w) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f_j(z)}{z-w} dz$$

$f(w) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(z)}{z-w} dz$ which implies $\bar{\partial} f = 0$

□

Theorem 1.8 (Montel's theorem) Suppose $\{f_i\} \subseteq A(\Omega)$ are uniformly bounded on each compact subset, then there is a subsequence f_{i_k} uniformly converges on compact subsets

Proof. For $K \subseteq \Omega$ compact, by Cauchy's estimate 1.6, f_j are Lipschitz with the same C_k, by Ascoli-Arzela theorem, f_j are equicontinuous, thus have convergent subsequence, and then use diagonal argument by exhaust Ω with compact subsets K

□

Theorem 1.9 (Riemann extension theorem) $E \subseteq \Omega$ is a discrete subset, $f \in A(\Omega \setminus E)$, and f is bounded around each point in E, then f can be extended to a unique $\tilde{f} \in A(\Omega)$

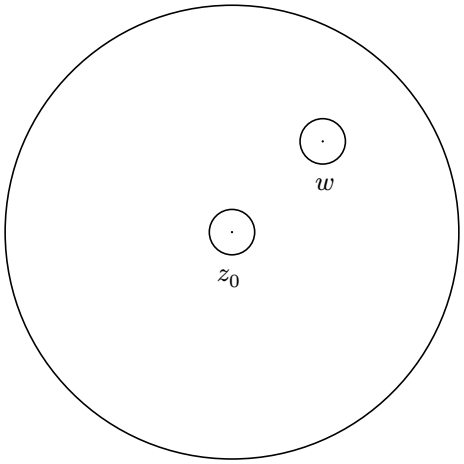
Proof. For $z_0 \in E$, suppose such $\tilde{f}$ exists, then by Cauchy's formula 1.4, for any $w \in D(z_0, r)$

$$\tilde{f}(w) = \frac{1}{2\pi i} \int_{\partial D(z_0, r)} \frac{f(z)}{z-w} dz$$

Thus we just take this as a definition, then

$$\begin{aligned}\tilde{f}(w) - f(w) &= \frac{1}{2\pi i}\int_{\partial D(z_0,r)} \frac{f(z)}{z-w}dz - \frac{1}{2\pi i}\int_{\partial D(w,\epsilon)} \frac{f(z)}{z-w}dz \\ &= \frac{1}{2\pi i}\int_{\partial D(z_0,\epsilon)} \frac{f(z)}{z-w}dz\end{aligned}$$

Which can be show to arbitrarily small as $\epsilon \to 0$



□

Theorem 1.10 If $\alpha = g(z)d\bar{z}$ is a smooth $(0,1)$-form on Ω, then there exists $u \in C^\infty(\Omega)$ such that $\bar{\partial}u = \alpha$

Proof. suppose such a u exists, then by Cauchy's formula 1.4

$$u(w) = \frac{1}{2\pi i}\iint_\Omega \frac{g(z)dz \wedge d\bar{z}}{z-w} + \frac{1}{2\pi i}\int_{\partial\Omega} \frac{u(z)}{z-w}dz$$

Since $\bar{\partial}\int_{\partial\Omega} \frac{u(z)}{z-w}dz = 0$. This motivates us to first assume α has compact support, and define

$$u(w) = \frac{1}{2\pi i}\iint_\Omega \frac{g(z)dz \wedge d\bar{z}}{z-w}$$

Then

$$u(w+\zeta) = \frac{1}{2\pi i}\iint_\Omega \frac{g(z)dz \wedge d\bar{z}}{(z-\zeta)-w} = \frac{1}{2\pi i}\iint_\Omega \frac{g(z+\zeta)dz \wedge d\bar{z}}{z-w}$$

Hence

$$\begin{aligned}\partial_{\bar{w}}u(w) &= \frac{1}{2\pi i}\iint_\Omega \frac{\partial_{\bar{z}}g(z)dz \wedge d\bar{z}}{z-w} \\ &= \frac{1}{2\pi i}\iint_\Omega \partial \log|\ z-w\ |^2 \wedge \bar{\partial}g \\ &= g(w)\end{aligned}$$

Therefore $\bar{\partial}u = \alpha$. In general, consider a compact exhaustion $\Omega = \bigcup_i K_i$, where $\hat{K}_i = K_i$, $K_i \subset\subset K_{i+1}^\circ$, ensured by Corollary 2.6, let χ_i be a cutoff function such that $\chi_i \equiv 1$ on K_i and supp $\chi_i \subseteq K_{i+1}^\circ$, then there exists f_i such that $\bar{\partial}f_i = \chi_i\alpha$, by Runge's theorem 2.2, there exists $h_i \in \mathcal{O}(K_i)$ such that $\|\ f_{i+1} - f_i - h_i\ \|_{K_i} < \frac{1}{2^i}$. Now define

$$u_N = f_1 + \sum_{k=1}^{N} (f_{k+1} - f_k - h_k) = f_{N+1} - \sum_{k=1}^{N} h_N$$

Converges uniformly on compact subsets to u, and $\partial u_N = \alpha$ on K_i for any $i \leq N$

□

2 Runge's theorem

Definition 2.1 $K \subseteq \Omega$ is compact, define

$$\mathcal{O}(K) = \{f \mid_K : f \text{ is holomorphic in a neighborhood of } K\}$$

Then for we have restriction map $\rho : \mathcal{O}(\Omega) \to \mathcal{O}(K)$, let $\| f \|_K = \max_{z \in K} | f(z) |$ to be the L^∞ norm

Theorem 2.2 (Runge's theorem) The following are equivalent

1. The image of ρ is dense
2. No connected component of $\Omega \setminus K$ is relatively compact in Ω
3. If $\xi \in \Omega \setminus K$, then there exists $f \in \mathcal{O}(\Omega)$ such that $| f(\xi) | > \| f \|_K$

Definition 2.3 For $K \subseteq \Omega$ compact, the *holomorphic convex hull* of K relative to Ω is

$$\hat{K} = \hat{K}_\Omega = \left\{ z \in \Omega : | f(z) | \leq \| f \|_K, \forall f \in \mathcal{O}(\Omega) \right\}$$

Clearly $K \subseteq \hat{K}$

Proposition 2.4

1. $\hat{K}$ is compact
2. $\| f \|_{\hat{K}} = \| f \|_K$ for all $f \in \mathcal{O}(\Omega)$
3. $\hat{\hat{K}} = \hat{K}$
4. If $\xi \in \Omega \setminus \hat{K}$, then there exists $f \in \mathcal{O}(\Omega)$ such that $| f(\xi) | > \| f \|_K$

Proof.

1. $\hat{K}$ is bounded by considering $f = z$. Suppose $z_i \in \hat{K}$ converges to ξ, if $\xi \in \Omega^c$, then $f = \frac{1}{z - \xi}$ will be unbounded on $\hat{K}$, thus $\xi \in \Omega$, but then for any $f \in \mathcal{O}(\Omega)$, $| f(\xi) | = \lim_{i \to \infty} | f(z_i) | \leq \| f \|_K$, thus $\xi \in \hat{K}$
2. By definition, $\| f \|_{\hat{K}} \leq \| f \|_K$, $\forall f \in \mathcal{O}(\Omega)$, and $\| f \|_K \leq \| f \|_{\hat{K}}$, $\forall f \in \mathcal{O}(\Omega)$ is obvious
3. $\hat{\hat{K}} = \left\{ z \in \Omega : | f(z) | \leq \| f \|_{\hat{K}} = \| f \|_K, \forall f \in \mathcal{O}(\Omega) \right\} = \hat{K}$
4. By definition

□

Example 2.5 K is the unit circle. If Ω is the anulus $\left\{ \frac{1}{2} < | z | < 2 \right\}$, then $\hat{K} = K$. If Ω is the disc $\{| z | < 2\}$, then $\hat{K} = \{| z | < 1\}$ is the unit disc. Just consider $f = z$ and $f = \frac{1}{z}$

Corollary 2.6 Any domain Ω has an exhaustion by compact sets $\widehat{K_i} = K_i$ such that

$$K_i \subset\subset K_{i+1}^\circ \subset K_{i+1} \subset\subset \Omega$$

Theorem 2.7 $\mathcal{U} = \{U_i\}$ is an open cover of Ω, then $H^1(\mathcal{U}, \mathcal{O}) = 0$

Proof. Let $\{\phi_i\}$ be a partion of unity. For any cocycle $\{g_{ij}\} \in Z^1(\mathcal{U}, \mathcal{O})$, consider $h_i = \sum_j \phi_j g_{ij}$, then

$$\begin{aligned} h_i - h_j &= \sum_k \phi_k g_{ik} - \sum_k \phi_k g_{jk} \\ &= \sum_k \phi_k (g_{ik} - g_{jk}) \\ &= \sum_k \phi_k g_{ij} \\ &= g_{ij} \end{aligned}$$

Hence $\bar{\partial} h_i - \bar{\partial} h_j = 0$, $\left\{\bar{\partial} h_i\right\}$ define a well-defined smooth $(0,1)$ form. By Theorem 1.10, there exist a holomorphic fuction u such that $\bar{\partial} u = \bar{\partial} h_i$, define $f_i = h_i - u$, then $\bar{\partial} f_i = 0$, i.e. $\{f_i\}$'s are holomorphic, and $g_{ij} = f_i - f_j$. In other words, $\left\{g_{ij}\right\}$ is the image of $\{f_i\} \in C^1(\mathcal{U}, \mathcal{O})$ under the coboundary map

□

Theorem 2.8 (Mittag-Leffler theorem) $\Omega \subseteq \mathbb{C}$ is an open set, $E \subseteq \Omega$ is a discrete subset, then there exists a meromorphic function f with prescribed principal parts on E

Proof. There exists and open cover $\mathcal{U} = \{U_i\}$ and $f_i \in \mathcal{M}(U_i)$ with the prescribed principal parts round each point of E, then $f_i - f_j \in \mathcal{O}(U_i \cap U_j)$ is a coycle, by Theorem 2.7, there exist holomorphic functions $\{g_i\}$ such that $f_i - f_j = g_i - g_j$ on $U_i \cap U_j$, then $f_i - g_i = f_j - g_j$ defines a global meromorphic function f such that $f - f_i = -g_i$ on U_i which is holomorphic

□

Theorem 2.9 (Weierstrass theorem) $E \subseteq \Omega$ is discrete, then

1. There is $f \in \mathcal{M}(\Omega)$ with arbitrary orders precisely at E
2. Any $f \in \mathcal{M}(\Omega)$ can be written as $f = g/h$ for $g, h \in \mathcal{O}(\Omega)$

Proof.

1. First take care of poles, and then multiply by $a_k(z - z_k)^{r_k}$ for each zero z_k, that converges
2.

□

Definition 2.10 Open subset $\Omega \subseteq \mathbb{C}^n$ is called a *domain of holomorphy* if for any $p \in \overline{\Omega} \setminus \Omega$, there is no holomorphic function g defined on an open set $U \ni p$ with $g = f$ on $U \cap \Omega$

Theorem 2.11 For any proper open subset $\Omega \subseteq \mathbb{C}$ is a domain of holomorphic

Proof. Suppose $p \in \partial\Omega$, $p \in U$ is a neighborhood, $g \in \mathcal{O}(U)$ such that $f = g$ on $\Omega \cap U$, then there exists $\{\xi_n\}$ discrete and converging to p. By Weierstrass theorem 2.9, there exists $f \in \mathcal{O}(\Omega)$ having exactly $\{\xi_i\}$ as zeros, but then g has to be identically zero, so is f which is a contradiction

□

3 Subharmonic functions

Definition 3.1 $\Omega \subseteq \mathbb{C}$ is a domain. $u : \Omega \to \mathbb{R} \cup \{-\infty\}$ is *upper semicontinuous* if for $y \in \mathbb{R}$ the set $\{u < y\}$ is open

Definition 3.2 An upper semicontinuous function u is *subharmonic* if is not identitcally $-\infty$, and for each $U \subset\subset \Omega$ and harmonic function h on $\overline{U}$ with $u \leq h$ on ∂U, we have $u \leq h$ for all $z \in U$

Example 3.3 If $u \in C^2(\Omega)$ and $\Delta u \geq 0$, then u is subharmonic

Theorem 3.4

1. If $\{u_i\}$ are subharmonic and $u = \sup u_i$ is finite and upper semicontinuous, then u is subharmonic
2. If $u_i \geq u_{i+1}$ are subharmonic, then $u = \lim u_i$ is subharmonic

Proof.

1. By definition
2. $\{u < y\} = \bigcup \{u_i < y\}$ is open, hence u is upper semicontinuous. Suppose $u \leq h$ on ∂U for some $U \subset\subset \Omega$ and harmonic function h. For any $\epsilon > 0$, consider

$$F_i = \{x \in \partial U \mid u_i(x) \geq h(x) + \epsilon\}$$

   are compact, thus $\bigcap F_i = \varnothing$ implies that a finite intersection is empty, hence $u \leq h + \epsilon$

□

Fact 3.5 If u is subharmonic on Ω, then $u \in L^1_{\text{loc}}(\Omega)$

Theorem 3.6 Subharmonic function u satisfies the sub-mean value property

$$u(z) \leq \frac{1}{2\pi} \int_0^{2\pi} u(z + re^{i\theta}) d\theta \tag{3.1}$$

For almost all r sufficiently small

Proof. u is integrable on circle of radius r about z for sufficiently small r, we can find continuous functions $h_n \geq u_n$ on the circle such that $h_n \to u$ in L^1, extend h_n to harmonic functions, then

$$u(z) \leq h_n(z) = \frac{1}{2\pi} \int_0^{2\pi} h_n(z + re^{i\theta}) d\theta \to \frac{1}{2\pi} \int_0^{2\pi} u(z + re^{i\theta}) d\theta$$

□

Proposition 3.7 Subharminoc functions satisfies $\Delta u \geq 0$ in the weak sense

$$\int_\Omega u \Delta \phi \geq 0, \forall \phi \in C_c^\infty(\Omega), \phi \geq 0$$

Proof. Multiply ϕ on both sides of (3.1) and integrate over Ω we get

$$\begin{aligned} \int_\Omega 2\pi u(z)\phi(z) d\mu &\leq \int_\Omega \phi(z) \int_0^{2\pi} u(z + re^{i\theta}) d\theta d\mu \\ &= \int_\Omega u(z) \int_0^{2\pi} \phi(z - re^{i\theta}) d\theta d\mu \end{aligned}$$

Then we get

$$
\begin{aligned}
0 &\leq \int_\Omega u(z) \int_0^{2\pi} \phi(z - re^{i\theta}) - \phi(z) d\theta d\mu \\
&= \int_\Omega u(z) \int_0^{2\pi} -\partial_z \phi(z) re^{i\theta} - \partial_{\bar{z}} \phi(z) re^{-i\theta} + \partial_z^2 \phi(z) r^2 e^{2i\theta} + \partial_{\bar{z}}^2 \phi(z) r^2 e^{-2i\theta} + 2\partial_z \partial_{\bar{z}} \phi(z) r^2 + O(r^3) d\theta d\mu \\
&= \int_\Omega u(z) \int_0^{2\pi} \frac{1}{2} \Delta \phi(z) r^2 + O(r^3) d\theta d\mu
\end{aligned}
$$

Divide $\frac{r^2}{2}$ and let $r \to 0$

□

Proposition 3.8 Subharmonicity is a local property, i.e. suppose u is upper semicontinuous on Ω, and locally subharmonic, then u is subharmonic on Ω

Proof. Suppose h is harmonic, $U \subset\subset \Omega$, $u \leq h$ on $\partial\Omega$, consider $v = u - h$, assume $\sup_U v = M > 0$, then by the upper semicontinuity, we know that $F = \{v = M\}$ is compact in U, there exists $z_0 \in \partial F$ obtains the least distance from ∂U, then for any small $r > 0$, F will miss an arc of positive measure if $\partial B(z_0, r)$, hence

$$
\frac{1}{2\pi} v(z_0 + re^{i\theta}) d\theta < M
$$

But this contradicts sub-mean value property

□

Example 3.9 If $f_1, \cdots, f_k \in \mathcal{O}(\Omega)$, not all zero, then $u = \log(\mid f_1 \mid^2 + \cdots + \mid f_k \mid^2)$ is subharmonic since $\log\mid f \mid$ is harmonic and $\Delta u \geq 0$

4 Almost complex structure

Definition 4.1 V is a real vector space, an *almost complex structure* is an endomorphism $J : V \to V$ such that $J^2 = -I$. Let $V^{1,0} \oplus V^{0,1} = V_{\mathbb{C}}$ be the $\pm i$ eigenspaces of J

Proposition 4.2 We can find basis such that $V \cong \mathbb{R}^{2n}$ such that $J = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$. For local coordinate (x_i, y_i) of a complex manifold, $\left\{\frac{\partial}{\partial x_i}, \frac{\partial}{\partial y_i}\right\}$ is such a basis, $\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_i}$ are the $\pm i$ eigenvectors. This motivates the definition of a real isomorphism $\rho : V \to V^{1,0}$, $v \mapsto \frac{1}{2}(v - iJv)$, then $\rho J = i\rho$. Suppose V, W both have almost complex structures, given an $\mathbb{R}$-linear map $T : V \to W$, let $\tilde{T} : V^{1,0} \to W^{1,0}$ be given by the commutative diagram

$$\begin{array}{ccc} V & \overset{T}{\to} & W \\ \underset{\rho}{\downarrow} & & \underset{\rho}{\downarrow} \\ V^{1,0} & \overset{\tilde{T}}{\to} & W^{1,0} \end{array}$$

$\tilde{T}$ is complex linear if $TJ = JT \Leftrightarrow \tilde{T}i = i\tilde{T}$. Alternatively, extend T to a map $V_{\mathbb{C}} \to W_{\mathbb{C}}$, and this conditions is exactly that this extension preserves $(1,0)$ and $(0,1)$ subspaces

Lemma 4.3 (Osgood's lemma) If $f : \Omega \to \mathbb{C}$ is continuous and holomorphic in each variable, then it is analytic

Proof. Iterate Cauchy's formula and use Fubini's theorem to write

$$f(z) = \left(\frac{1}{2\pi i}\right)^n \int_{w_i \in \Delta(z_i, r_i)} \frac{f(w)dw}{(w_1 - z_1)\cdots(w_n - z_n)}$$

Then

$$\frac{1}{(w_1 - z_1)\cdots(w_n - z_n)} = \sum_I \frac{(z - \xi)^I}{(w - \xi)^I}$$

Then a convergent power series expression follows, with

$$c_I \left(\frac{1}{2\pi i}\right)^n \int_{w \in \Delta(z,r)} \frac{f(w)dw}{(w_1 - z_1)^{i_1+1}\cdots(w_n - z_n)^{i_n+1}}$$

The *total order* of an analytic function f at ξ is the smallest value of $\mid I \mid$ for which $c_I \neq 0$

□

Definition 4.4 A set $E \subseteq \Omega$ is called *thin* if for every $\xi \in E$ there is a polydisk $\Delta(\xi, r) \subset\subset \Omega$ and $g \in A(\Delta(\xi, r))$ such that $E \cap \Delta(\xi, r) \subseteq Z(g)$. Note that for $n = 1$, this is equivalent to discrete

Theorem 4.5 (Riemann extension theorem) If $f \in A(\Omega \setminus E)$ where E is a thin set, and f is locally bounded on Ω, then there exists $\tilde{f} \in A(\Omega)$ such that $\tilde{f} = f$ on the complement of E

Proof. Let k be the total order of g at ξ. By an application of Rouché's theorem (and after modifying r and a change of variables), we can assume that for each $z_1, \cdots, z_{n-1}$ the function $z_n \mapsto g(z_1, \cdots, z_{n-1}, z_n)$ has exactly k zeros and none on the boundary

□

In higher dimensions, to solve $\bar{\partial}$ equation, there must be a *integrability condition*. Indeed, if we can solve the equation, then $0 = \bar{\partial}^2 u = \bar{\partial}\alpha$, i.e. we require α to be $\bar{\partial}$ closed

Proposition 4.6 Let $n \geq 2$. If α is a smooth compactly supported $(0,1)$ form on $\mathbb{C}^n$ with $\bar{\partial}\alpha = 0$, then there is a $u \in C_c^\infty$, with $\bar{\partial}u = \alpha$

Proof.

□

Corollary 4.7 (Hartogs theorem) Let $K \subseteq \Omega$ be compact with $\Omega \setminus K$ connected. If $f \in A(\Omega \setminus K)$, there exists $\tilde{f} \in A(\Omega)$ that is equal to f on the complement of K

Proof. Let $\phi \in C_c^\infty(\Omega)$ be $\equiv 1$ in a neighborhood of K, let $\alpha = \bar{\partial}((1-\phi)f)$. Then α is $\bar{\partial}$-closed and compactly supported. Hence, there is $u \in C_c^\infty(\mathbb{C}^n)$ with $\bar{\partial} u = \alpha$. Then let $\tilde{f} = (1-\phi)f - u$, $\tilde{f} \in A(\Omega)$, since u is compactly supported, $\tilde{f} = f$ on $\Omega \setminus K$

□

Note. The assumption that $\Omega \setminus K$ is connected is necessary. For example, let $K \subseteq B(0,1) = \{| z | < 1\}$ be the set where $| z | = \frac{1}{2}$, and take

$$f(z) = \begin{cases} z_n \text{ if } 1/2 < | z | < 1 \\ 0 \text{ if } | z | < 1/2 \end{cases}$$

Then there is no holomorphic extension to $B(0,1)$

Proposition 4.8 If α is a smooth $\bar{\partial}$-closed $(0,1)$ form on a polydisk $\Delta = \Delta(0,r)$, then $\alpha = \bar{\partial} u$ for some $u \in C^\infty(\Delta)$

Proof. Just like in the one variable case, exhaust Δ by nested closed polydiscs K_i. Use cut-off functions to find u_i, $\bar{\partial} u_i$ in a neighborhood of K_i. Then $u_{i+1} - u_i$ is holomorphic in a neighborhood of K_i. Now by the power series expansion, there is a polynomial p_i such that $\| u_{i+1} - u_i - p_i \|_{K_i} < 2^{-i}$. The rest follows as in the proof of the one variable case

□

Note. We heavily used the geometric properties of the polydisc

Corollary 4.9 (Cousin theorem) $\mathcal{U} = \{u_i\}$ is an open cover of polydisc Δ, then $H^1(\Delta, \mathcal{U}) = 0$

Theorem 4.10 If $\alpha \in C^\infty_{(p,q)}(\Delta)$, $q \geq 1$, $\bar{\partial}\alpha = 0$. Then $\alpha = \bar{\partial} u$ for some $u \in C^\infty_{(p,q-1)}(\Delta)$

Remark 4.11 This states that the Dolbeault cohomology groups $H^{p,q}_{\bar{\partial}}(\Delta) = 0$

Proof. Induct on $k = 1, \cdots, n$, the smallest integer such that α only involves $d\bar{z}_1, \cdots, d\bar{z}_k$. If $k = 1$, then $q = 1$ and we have already proven the result. Suppose the result is true for $k-1$. Write $\alpha = \omega \wedge d\bar{z}_k + \beta$, where ω and β only involve $d\bar{z}_1, \cdots, d\bar{z}_{k-1}$. We have $0 = \bar{\partial}\alpha = \bar{\partial}\omega \wedge d\bar{z}_k + \bar{\partial}\beta$. This implies both ω, β are holomorphic in the variables $z_{k+1}, \cdots, z_n$. Apply the one variable solution to find μ, $\bar{\partial}\mu = \omega \wedge d\bar{z}_k + \sigma$, here σ only involves $d\bar{z}_1, \cdots, d\bar{z}_{k-1}$. Now $\alpha - \bar{\partial} u = \beta - \sigma$ is $\bar{\partial}$-closed. By induction, we can write $\beta - \sigma = \bar{\partial} v$, and so we set $u = v + \mu$

□

Example 4.12 Let $\Omega \subseteq \mathbb{C}^2$ be a domain. For $\xi \in \Omega$, let $\Omega^* = \Omega \setminus \{\xi\}$. Then $H^{0,1}_{\bar{\partial}}(\Omega^*) \neq \{0\}$

Proof. Without loss of generality assume $\xi = (0,0)$. Consider the $(0,1)$-form

$$\omega = \frac{1}{r^4}(-\bar{z}_2 d\bar{z}_1 + \bar{z}_1 d\bar{z}_2) = \bar{\partial}\left(\frac{\bar{z}_2}{z_1 r^2}\right)$$

Clearly, ω is smooth on Ω^*, and $\bar{\partial}\omega = 0$. Suppose $\omega = \bar{\partial} u$ for $u \in C^\infty(\Omega^*)$. Then $f(z_1, z_2) = z_1 u - \frac{\bar{z}_2}{r^2}$ is holomorphic on $\Omega^* \setminus \{z_1 = 0\}$, and it is locally bounded on Ω^*. By Riemann extension, it is holomorphic on Ω^*. By Hartogs, it extends to Ω. But for $z_2 \neq 0$ we clearly have $f(0, z_2) = -\frac{1}{z_2}$, contradiction

□

Proposition 4.13 $K \subseteq \Omega$ is compact

1. $\hat{K}_\Omega$ is closed in Ω
2. $\hat{K}_\Omega$ is not necessarily closed in $\mathbb{C}^n$. E.g. if $n \geq 2$, let $\Omega = \mathbb{B}^n \setminus \{0\}$, $K = \{| \ z \ | = 1/2\}$. Then by Hartogs' theorem, $\hat{K}_\Omega = \mathbb{B}^n_{1/2} \setminus \{0\}$
3. $\hat{K}_\Omega \subseteq \mathcal{C}(K)$, the closed convex hull of K. In particular, $\hat{K}_\Omega$ is bounded

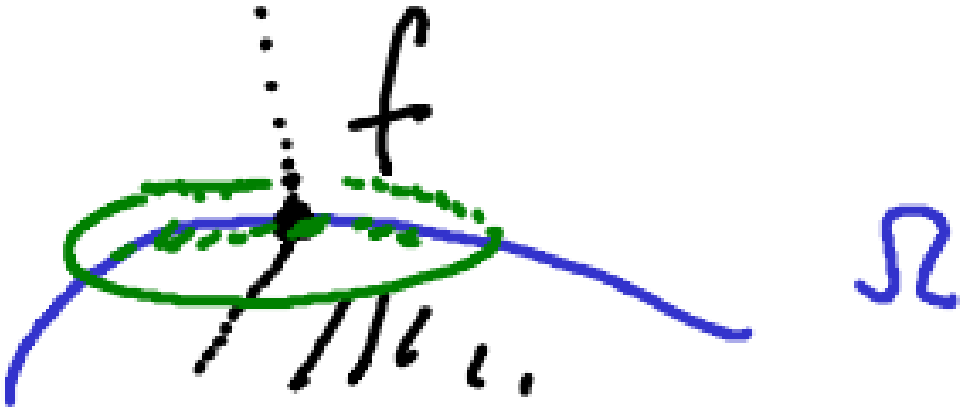
Proof. Let $w \notin \mathcal{C}(K)$, $z_0 \in \mathcal{C}(K)$ minimizes distance to w, let $\xi \in (\mathbb{C}^n)^*$ define a supporting hyperplane for $\mathcal{C}(K)$ so that $\mathcal{C}(K) \subseteq \operatorname{Re} \langle \xi, z \rangle \leq 0$ and $\operatorname{Re} \langle \xi, w \rangle \geq 0$. Let $f(z) = \exp\langle \xi, z \rangle$, $| \ f(z) \ | = \exp \operatorname{Re} \langle \xi, z \rangle$ which violates the definition, so $w \notin \hat{K}_\Omega$

□

Definition 4.14 A domain $\Omega \subseteq \mathbb{C}^n$ is *holomorphically convex* if for every compact $K \subseteq \Omega$, $\hat{K}_\Omega$ is compact. If Ω is convex, then it is holomorphically convex. If $n = 1$, all domains are holomorphically convex. The previous counter-example shows this is not true if $n \geq 2$

Proposition 4.15 $\Omega \subseteq \mathbb{C}^n$ is holomorphically convex $\Leftrightarrow$ every discrete, infinite set $\{z_j\} \subseteq \Omega$ there is $f \in A(\Omega)$ with $| \ f(z_j) \ |$ unbounded

Proof. $\Leftarrow$: If $\hat{K}_\Omega$ is not compact there is a discrete infinite subset $\{z_j\} \subseteq \hat{K}$. But then $| \ f(z_j) \ | \leq \| \ f \ \|_K$, $\forall j, f \in A(\Omega)$. This contradicts the existence of $f \in A(\Omega)$ where $| \ f(z_j) \ |$ is unbounded



$\Rightarrow$: Exhaust Ω by nested compact sets K_j, $\hat{K}_j = K_j$. We may assume $z_j \in K_{j+1} \setminus K_j$. We can find $f_j \in A(\Omega)$ such that $f_j(z_j) = 1$, $\| \ f_j \ \|_{K_j} < 1$, by taking power, $\| \ f_j \ \|_{K_j}$ can actually be arbitrarily small. Let $g_j \in A(\Omega)$ be such that $g_j(z_j) = 1$, $g_j(z_j) = 0$ for $i < j$. Now define λ_j by

$$\lambda_j = j - \sum_{i=1}^{j-1} \lambda_i g_i f_i(z_j)$$

Assume $\| \ \lambda_j g_j f_j \ \|_{K_j} < 2^{-j}$. Now let $f(z) = \sum_{i=1}^{\infty} \lambda_i g_i f_i(z)$. This converges uniformly on compact sets, and so $f \in A(\Omega)$. Finally

$$f(z_j) = \sum_{i=1}^{j} \lambda_i g_i f_i(z_j) = \lambda_j g_j f_j(z_j) + \sum_{i=1}^{j-1} \lambda_i g_i f_i(z_j) = j$$

□

Definition 4.16 $\Omega \subseteq \mathbb{C}^n$ is called a *domain of holomorphy* if there is $f \in A(\Omega)$ such that for any $p \in \overline{\Omega} \setminus \Omega$ and any Ω' about p, there is no $g \in A(\Omega')$ such that $g = f$ on $\Omega' \cap \Omega$

Theorem 4.17 $\Omega \subseteq \mathbb{C}^n$ is holomorphically convex $\Leftrightarrow$ it is a domain of holomorphy

Corollary 4.18 A convex domain in $\mathbb{C}^n$ is a domain of holomorphy

Proof. $\Rightarrow$ is similar to the one variable case. $\Leftarrow$ is a theorem of Oka (this will be generalized)
$\Rightarrow$: Fix a polydisc Δ about the origin. For $\xi \in \Omega$, let $\Delta_\xi = \xi + r\Delta$, where r is the supremum such that $\xi + r\Delta \subseteq \Omega$. Let $E \subseteq \Omega$ be countable dense. Let $\{\xi_j\}$ be a sequence containing every point of E infinitely may times. Write $\Omega = \bigcup K_j$. Since $\hat{K}_j \subset\subset \Omega$, $\exists z_j \in \Delta_{\xi_j}$ with $z_j \notin \hat{K}_j$. Choose $f_j \in A(\Omega)$, $f_j(z_j) = 1$, $\| f_j \|_{K_j} < 2^{-j}$. Set $f(z) = \prod(1 - f_j)^j$. Then f converges uniformly on compact sets, so $f \in A(\Omega)$. Now f has zeros of order $\geq j$ at z_j. Any continuation of f would have a zero of infinite order
$\Rightarrow$: Let $d(z) = \sup_{\Delta(z,r)\subseteq\Omega} r$, $d(K) = \inf_{z\in K} d(z)$. Claim $d(\hat{K}) = d(K) > 0$. This will imply $\hat{K} \subset\subset \Omega$. Let $f \in A(\Omega)$ so that the radius of convergence at z is $d(z)$, let $\delta < d(K)$, $K_\delta = \bigcup_{w\in K} \overline{\Delta(w, \delta)}$. By Cauchy estimates: $\| D^I f \|_K \leq \frac{I!}{\delta^{|I|}} \| f \|_{K_\delta}$. But $D^I f \in A(\Omega)$, so for $z \in \hat{K}$, $| D^I f(z) | \leq \| D^I f \|_K \leq \frac{I!}{\delta^{|I|}} \| f \|_{K_\delta}$. This implies that the radius of convergence at $z \in \hat{K}$ is at least δ, i.e. $d(z) \geq \delta$, and so $d(\hat{K}) \geq d(K)$. Since $K \subseteq \hat{K}$, the other inequality is trivial

□

Proposition 4.19 If $\{\Omega_\alpha\}_{\alpha\in I}$ are domains of holomorphy in $\mathbb{C}^n$, then the interior Ω of $\bigcap_{\alpha\in I} \Omega_\alpha$ is also a domain of holomorphy

Proof. $K \subseteq \Omega$ is compact. For each $\alpha \in I$, $K \subseteq \Omega \subseteq \Omega_\alpha$, which implies $\hat{K}_\Omega \subseteq \hat{K}_{\Omega_\alpha}$. This implies $d_{\Omega_\alpha}(\hat{K}_{\Omega_\alpha}) \leq d_{\Omega_\alpha}(\hat{K}_\Omega)$, for all α. Since Ω_α is holomorphically convex, $d_{\Omega_\alpha}(\hat{K}_{\Omega_\alpha}) = d_{\Omega_\alpha}(K)$. Hence $d_\Omega(K) \leq d_{\Omega_\alpha}(K) \leq d_{\Omega_\alpha}(\hat{K}_\Omega)$. Finally, this implies $d_\Omega(K) \leq d_\Omega(\hat{K}_\Omega)$. As before, we conclude that $d_\Omega(K) = d(\hat{K}_\Omega)$, and so $\hat{K}_\Omega$ is compact, so Ω is holomorphically convex

□

Claim. Suppose Ω is a domain of holomorphy. Let $f_1, \cdots, f_N \in A(\Omega)$, and define

$$\Omega_c = \{z \in \Omega \mid | f_j(z) | < c, j = 1, \cdots, N\}$$

Then Ω_c is also a domain of holomorphy

Proof. Let $K \subseteq \Omega_c$. Let $z \in \hat{K}_\Omega$. Then in particular, for any $j = 1, \cdots, N$, $| f_j(z) | \leq \| f_j \|_K < c$. So $z \in \Omega_c$. Now $\hat{K}_{\Omega_c} \subseteq \hat{K}_\Omega \subseteq \Omega$ and so $\hat{K}_{\Omega_c}$ is compact

□

Claim. Let $u : \Omega \subseteq \mathbb{C}^n \to \mathbb{C}^m$ be holomorphic, with Ω a domain of holomorphy. If $\Omega' \subseteq \mathbb{C}^m$ is a domain of holomorphy, then so is $\tilde{\Omega} = u^{-1}(\Omega')$

Proof. Let $K \subseteq \tilde{\Omega} \subseteq \Omega$ be compact. Since $\hat{K}_{\tilde{\Omega}} \subseteq \hat{K}_\Omega \subseteq \Omega$, it suffices to show $\hat{K}_{\tilde{\Omega}}$ is closed in Ω. Let $z_j \to z \in \Omega$, $z_j \hat{K}_{\tilde{\Omega}}$. Notice that $u(\hat{K}_{\tilde{\Omega}}) \subseteq \widehat{u(K)}_{\Omega'}$. Hence $u(z) \in \Omega'$, and so $z \in \tilde{\Omega}$

□

Lemma 4.20 Let $\Omega \subseteq \mathbb{C}^n$ be a domain of holomorphy, and $K \subseteq \Omega$. Suppose $f \in A(\Omega)$ is such that $| f(z) | \leq d(z)$ for all $z \in K$, then $| f(\xi) | \leq d(\xi)$ for all $\xi \in \hat{K}_\Omega$

Proof. We first claim that if $u \in A(\Omega)$, then the power series expansion of u at $\xi \in \hat{K}_\Omega$ converges on $\Delta(\xi, | f(\xi) |)$. This will prove the Lemma, because we can take u to be teh function with no analytic continuation beyond Ω

Proof of the claim: Let $0 < \delta < 1$, as before, the Cauchy estimates provide for some constant M that

$$\mid D^I u(z) \mid \frac{(\delta \mid f(z) \mid)^{\mid I \mid}}{I!} \leq M, \forall z \in K$$

Now $D^I u(z) f(z)^{\mid I \mid} \in A(\Omega)$, so the same estimate holds on $\hat{K}_\Omega$. This means the radius of convergence at $\xi \in \hat{K}_\Omega$ is a t least $\delta \mid f(\xi) \mid$. Since δ was arbitrary, this proves the claim

□

Fundamental consequence: Let $D \subset\subset \Omega$ be a 1-dimensional disc

1. Suppose f is a polynomial in one variable such that $-\log d(z) \leq \text{Re } f(z)$, for $z \in \partial D$
2. Let f be the restriction of $F \in A(\Omega)$. Then $\mid e^{-F(z)} \mid \leq d(z)$, $z \in \partial D$
3. By the maximum principle, $D \subseteq \widehat{\partial D_\Omega}$
4. From the Lemma, we have $\mid e^{-F(z)} \mid \leq d(z)$, $z \in \partial D$
5. This in turn implies $-\log d(z) \leq \text{Re } f$ on D

Approximating harmonic functions by polynomials, we conclude that $u = -\log d$ is subharmonic on any complex line in Ω

5 Hartogs theorem

Theorem 5.1

6 Pseudoconvexity

Definition 6.1 An upper semicontinuous function $\phi : \Omega \subseteq \mathbb{C}^n \to [-\infty, \infty)$ is *plurisubharmonic* if the restriction of ϕ to every complex line $L \cap \Omega$, $L \cong \mathbb{C}$, is subharmonic. Let $P(\Omega)$ be the set of plurisubharmonic (psh) functions on Ω

Proposition 6.2 $\phi \in C^2(\Omega)$ is psh $\Leftrightarrow$ for all $\xi \in \mathbb{C}^n$ and all $z \in \Omega$, the complex Hessian is positive semidefinite

$$\sum_{i,j=1}^{n} \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(z) \xi_i \bar{\xi}_j \geq 0$$

ϕ is strictly psh if $>$ holds for every $\xi \neq 0$

Remark 6.3 A real $(1,1)$ form on Ω can be written as

$$\omega(z) = i \sum_{i,j=1}^{n} g_{i\bar{j}}(z) dz_i \wedge d\bar{z}_j$$

where $g_{i\bar{j}}$ is a Hermitian matrix. We say that $\omega \geq 0$(resp. $\omega > 0$) if $(g_{i\bar{j}}(z))$ is positive semidefinite(resp. positive definite) for every $z \in \Omega$. This means that for each $\xi \in \mathbb{C}^n$, $\xi \neq 0$

$$\sum_{i,j=1}^{n} g_{i\bar{j}}(z) \xi_i \bar{\xi}_j \geq 0 (\text{resp. } \omega > 0)$$

In the case $\omega > 0$, $g_{i\bar{j}}$ defines a Hermitian metric on Ω, and ω is its associate Kähler form

Proof. A line $\jmath : L \hookrightarrow \mathbb{C}^n$ is given by a choice $\xi \neq 0$ in $\mathbb{C}^n$, so that $\jmath(\tau) = z_0 + \tau\xi$, then

$$\jmath^*(dz_i) = \xi_i d\tau, \jmath^*(d\bar{z}_i) = \bar{\xi}_i d\bar{\tau}$$

$$\begin{aligned} \jmath^*(i\partial\bar{\partial}\phi) &= \left(\sum_{i,j=1}^{n} \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(z) \xi_i \bar{\xi}_j \right) i d\tau \wedge d\bar{\tau} \\ &= \left(\sum_{i,j=1}^{n} \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(z) \xi_i \bar{\xi}_j \right) 2 d\mu \end{aligned}$$

On the other hand

$$\jmath^*(i\partial\bar{\partial}\phi) = i\partial_z \bar{\partial}_z (\phi \circ \jmath) = \Delta(\phi \circ \jmath) 2 d\mu$$

□

Definition 6.4 A domain $\Omega \subseteq \mathbb{C}^n$ is *pseudoconvex* if there exists a continuous psh exhaustion function ϕ, i.e.

$$\Omega_c = \{z \in \Omega \mid \phi(z) < c\} \subset\subset \Omega$$

For every $c \in \mathbb{R}$

Fact 6.5 (Richberg) If Ω is pseudoconvex, there is a C^∞ strictly psh exhaustion function on Ω (see Demailly's book)

Theorem 6.6 $\Omega \subseteq \Omega$ is a domain of holomorphy iff it is pseudoconvex

Proof. Recall $d(z) = \sup_{\Delta(z,r) \subseteq \Omega} r$. $\Rightarrow$: We have shown that $-\log d(z)$ is psh. It is also continuous. We claim that $u(z) = \mid z \mid^2 - \log d(z)$ does the job Closedness: If $z_i \to w \in \overline{\Omega} \setminus \Omega$, then $d(z_i) \to 0$, so u diverges Boundedness: Fix any $w \in \overline{\Omega} \setminus \Omega$, then

$$d(z) \leq \mid z - w \mid \leq \mid z \mid + \mid w \mid$$

so for $\mid z \mid$ large

$$\log d(z) \leq 2\log\mid z \mid \leq \frac{1}{2} \mid z \mid$$

This means a bound on u implies a bound on $\mid z \mid$

□

Example 6.7

1. Geometrically convex sets are pseudoconvex(e.g. balls and polydisks)
2. If $\{\Omega_\alpha\}$ are pseudoconvex, then the interior Ω of $\bigcap \Omega_\alpha$ is pseudoconvex
3. Annuli or punctured domains are not pseudoconvex
4. Let $\Omega \subseteq \mathbb{C}^n$ be pseudoconvex, $f_1, \cdots, f_k \in A(\Omega)$, then $\tilde{\Omega} = \Omega \setminus V(f_1) \cup \cdots \cup V(f_k)$ is pseudoconvex. Indeed, if ϕ is the psh exhaustion function on Ω, take $\tilde{\phi} = \phi - \log\mid f_1 \mid - \cdots - \log\mid f_k \mid$ on $\tilde{\Omega}$

Proposition 6.8 Suppose $\Omega \subseteq \mathbb{C}^n$ is pseudoconvex. Then $-\log d(z)$ is psh

Proof. $D \subset\subset \Omega$ is a disc, f on D, $F \in A(\Omega)$ restricts to f, suppose $-\log d(z) \leq \text{Re } f(z)$, $z \in \partial D$, or equivalently $d(z) \geq \mid e^{-f(z)} \mid$, $z \in \partial D$. We want to show this holds in D. Fix $w \in \Delta(0,1)$. Let

$$K = \left\{z + \lambda w e^{-f(z)} \mid z \in \partial D, 0 \leq \lambda \leq 1\right\}$$

Then $K \subseteq \Omega$

$$\Lambda = \left\{\lambda \in [0,1] \mid z + \lambda' w e^{-f(z)} \in \Omega, \forall z \in D, 0 \leq \lambda' \leq \lambda\right\}$$

Notice that $\Lambda \neq \varnothing$, since $0 \in \Lambda$. We want show that $\Lambda = [0,1]$. Λ is clearly open. Suppose $\lambda_i \nearrow c$, $\lambda_i \in \Lambda$, let ϕ be a continuous psh exhasution function on Ω, then for each j, $z \in D$, $\phi(z + \lambda_j w e^{-f(z)}) \leq \sup_K \phi$, but since this is a compact set, $c \in \Lambda$

□

Pseudoconvexity is a property of the boundary of Ω

Proposition 6.9 $\Omega \subseteq \mathbb{C}^n$. Suppose that for every $\xi \in \overline{\Omega}$ there is an open set $\xi \in U$ such that $U \cap \Omega$ is pseudoconvex. Then Ω is a pseudoconex

Proof. Let $\xi \in \partial\Omega$, set $\tilde{\Omega} = U \cap \Omega$. For z sufficiently close to ξ, $d(z) = d_\Omega(z) = d_{\tilde{\Omega}}(z)$, so $-\log d(z)$ is psh in a neighborhood of $\partial\Omega$(say, $\Omega \setminus F$ for smote closed F).Find a smooth proper psh function ψ on $\mathbb{C}^n$ such that $\phi(z) > -\log d(z)$ for $z \in F$. Now let $\phi(z) = \max\{\psi(z), -\log d(z)\}$. Then ϕ is a continuous psh exhaustion function

□

Definition 6.10 $\Omega \subseteq \mathbb{C}^n$ have a C^2 boundary. In a neighborhood U of $z_0 \in \partial\Omega$ we can find a C^2 defining function $\rho : U \to \mathbb{R}$, i.e.

$$\Omega \cap U = \{z \in U \mid \rho(z) < 0\}, \nabla\rho \neq 0 \text{ on } \partial\Omega \cap U$$

The *Levi form* L_{z_0} at the point z_0 is the quadratic form Hess (ρ) restricted to $V_{z_0} = T_{z_0}\partial\Omega \cap J(T_{z_0}\partial\Omega)$. Alternatively, let $\xi \in \mathbb{C}^n$ satisfy $\sum_{i=1}^n \frac{\partial\rho}{\partial z_i}\xi_i = 0$. Then we define

$$L(\xi) = \sum_{i,j=1}^n \frac{\partial^2\rho}{\partial z_j \partial\bar{z}_j}(z_0)\xi_i\bar{\xi}_j$$

Here, if ξ is the vector corresponding to v then $L(v) = L(\xi)$

Lemma 6.11 Let $z, w \in \Omega$, $\xi \in \Delta(0,r)$ such that $z = w + \xi$. Then $d(z) \geq d(w) - r$

Proof. Let η be in some polydisk about 0, such that $z+\eta \in \partial\Omega$, and $d(z) = \max|\ \eta_i\ |$. Then $w + \xi + \eta \in \partial\Omega$. This implies

$$d(w) \leq \max_j|\ (\xi+\eta)_j\ | \leq \max_j|\ \xi_j\ | + \max_j|\ \eta_j\ | \leq r + d(z)$$

□

Proposition 6.12 Ω is pseudoconvex $\Leftrightarrow$ the Levi form is everywhere positive semidefinite on $\partial\Omega$

Proof. $\Rightarrow$: $\rho(z) = \begin{cases} -d_\Omega(z) & z\in\Omega \\ 0 & z\in\partial\Omega \\ -d_{\overline{\Omega}^c}(z) & z\in\overline{\Omega}^c \end{cases}$, then ρ is C^2. The function $\phi = -\log d$ is C^2 and psh

$$\frac{\partial^2\phi}{\partial z_i \partial\bar{z}_j} = -\frac{1}{d(z)}\frac{\partial^2 d}{\partial z_i\partial\bar{z}_j} + \frac{1}{d(z)^2}\frac{\partial d(z)}{\partial z_i}\frac{\partial d(z)}{\partial\bar{z}_i}$$

So for $z \in \Omega$

$$0 \leq \sum_{i,j=1}^n \frac{\partial^2\phi}{\partial z_i\partial\bar{z}_j}(z)\xi_i\bar{\xi}_j = \sum_{i,j=1}^n \frac{1}{d(z)}\frac{\partial^2 d}{\partial z_i\partial\bar{z}_j}$$

Now let $z \to \partial\Omega$

$\Leftarrow$: Suppose $c = \frac{\partial^2}{\partial\tau\partial\bar{\tau}}\log d(z_0+\tau w_0) > 0$. $\log d(z_0+\tau w_0) = \log d(z_0) + \text{Re}\ (A\tau + B\tau^2) + c\ |\ \tau\ |^2 + o(|\ \tau\ |^2)$. Choose $\xi_0 \in \partial\Delta(0, d(z_0))$ such that $z_0+\xi_0 \in \partial\Omega$, $\max_i|\ \xi_{0,i}\ | = d(z_0)$. Let $z(\tau) = z_0 + \tau w_0 + \xi_0\exp(A\tau + B\tau^2)$. By Lemma 6.11

$$\begin{aligned} d(z(\tau)) &\geq d(z_0+\tau w_0) - d(z_0)\ |\exp(A\tau+B\tau^2)\ | \\ &\geq |\exp(A\tau+B\tau^2)\ |\ (e^{c\ |\ \tau\ |^2/2} - 1) \end{aligned}$$

Now $d(z(0)) = 0$. The inequalitity implies

$$\frac{\partial}{\partial\tau}d(z(\tau))|_{\tau=0} = 0, \frac{\partial^2}{\partial\tau\partial\bar{\tau}}d(z(\tau))|_{\tau=0} > 0$$

In other words

$$\sum_{i=1}^n \frac{\partial\rho}{\partial z_i}z_i'(0) = 0, \sum_{i,j=1}^n \frac{\partial^2\rho}{\partial z_i\partial\bar{z}_j}z_i'(0)\bar{z}_j'(0) < 0$$

This contradicts $L_{z(0)} \geq 0$

□

7 Hörmander's L^2 estimate

Definition 7.1 H_1, H_2 are complex Hilbert space, $T : H_1 \to H_2$ is an *unbounded operator*, if it is a linear map defined on some linear subspace $D(T) \leq H_1$ called the domain of T. T is *densely defined* if $D(T)$ is dense in H_1. T is *closed* if the graph $\text{Gr}\ (T) = \{(x, Tx) \in H_2 \times H_2 \mid x \in D(T)\}$ is closed. T has *closed range* if $R(T) = \{Tx \in H_2 \mid x \in D(T)\}$ is closed in H_2. Write $N(T) = \ker T$

Definition 7.2 $T : H_1 \to H_2$ is a densely defined unbounded operator, its adjoint $T^* : H_2 \to H_1$ is defined as an unbounded operator as follows

- $D(T^*)$ consists of $y \in H_2$ such that the funcitonal $\langle T(-), y\rangle : D(T) \to \mathbb{C}$ is continuous
- By the Hahn-Banach theorem, $\langle T(-), y\rangle$ extends to a linear functional on H_1
- By the Riesz representation theorem and denseness, there is a vector $T^*y \in H_1$ such that $\langle Tx, y\rangle = \langle x, T^*y\rangle$

Proposition 7.3 If T is densely defined, then T^* is closed

Proof. Let $y_j \in D(T^*)$, $y_j \to y$, and $x_j = T^*y_j \to x$. We need to show $y \in D(T^*)$ and $x = T^*y$. Fix $u \in D(T)$. Then

$$\mid u \mid \mid x \mid \geq \langle u, x\rangle = \lim_j \langle u, x_j\rangle = \lim_j \langle u, T^*y_j\rangle = \lim_j \langle Tu, y_j\rangle = \langle Tu, y\rangle$$

So the map $u \mapsto \langle Tu, y\rangle$ is bounded on $D(T)$ by $\mid x \mid$. This implies $y \in D(T^*)$ and $x = T^*y$

□

Fact 7.4 If T, T^* are densely defined then T is closed, and $(T^*)^* = T$

Lemma 7.5 If T is closed and densely defined, then $\text{Gr}\ (T^*) = \text{Gr}\ (-T)^\perp$ in $H_1 \times H_2$

Proof. We have inclusion $\subseteq$ since

$$\langle (T^*y, y), (x, -Tx)\rangle = \langle T^*y, x\rangle - \langle y, Tx\rangle = 0$$

Now if $\langle (x, y), (u, -Tu)\rangle = \langle x, u\rangle - \langle y, Tu\rangle = 0$ for all $u \in D(T)$, then $u \mapsto \langle Tu, y\rangle = \langle u, x\rangle$ is bounded on $D(T)$, so $y \in D(T^*)$, and $x = T^*y$

□

Theorem 7.6 If T is closed and densely defined, then so is T^*. Moreover, $N(T^*) = R(T)^\perp$ and $N(T) = \overline{R(T^*)^\perp}$

Note. $(V^\perp)^\perp = \overline{V}$

Proof. By Lemma 7.5, any $(u, v) \in H_1 \times H_2$ can be written as

$$(u, v) = (x, -Tx) + (T^*y, y), x \in D(T), y \in D(T^*)$$

Taking $u = 0$, then $v = y + TT^*y$. This implies $\langle v, y\rangle = \mid y \mid^2 + \mid T^*y \mid^2$. If $v \in D(T^*)^\perp$, then $y = 0$, and so $v = 0$. Hence $D(T^*)$ must be dense. $N(T^*) = R(T)^\perp$ follows form $\langle Tx, y\rangle = \langle x, T^*y\rangle$

□

Proposition 7.7 Let $T : H_1 \to H_2$ be closed and densely defined. The following are equivalent

1. $R(T)$ is closed
2. $\exists C$ such that $\mid x \mid \leq C \mid Tx \mid$ for all $x \in D(T) \cap R(T^*)$
3. $R(T^*)$ is closed
4. $\exists C$ such that $\mid y \mid \leq C \mid T^*y \mid$ for all $y \in D(T^*) \cap R(T)$

Proof. 2.⇒1.: Suppose $Tx_j \to y$, then x_j converges, say to x, $(x_j, Tx_j) \to (x, y)$
To show 1.⇒2., recall $N(T) = R(T^*)^\perp$. Hence T is continuous and 1-1 from $D(T) \cap R(T^*)$ onto the closed subspace $R(T)$. Hence the inverse is continuous by the closed graph theorem. This proves 2.
3.⇔4.
2.⇒4.:

$$| \langle Tx, y\rangle | = | \langle x, T^*y\rangle | \leq | x | \, | T^*y | \leq C \, | Tx | \, | T^*y |$$

So $| \langle z, y\rangle | \leq C \, | T^*y | \, | z |$ for $z \in R(T)$, $y \in D(T^*)$

□

Definition 7.8 Now consider densely defined closed unbounded operators $H_1 \overset{T}{\to} H_2 \overset{S}{\to} H_3$ satisfying $S \circ T = 0$. The *harmonic elements* are

$$\mathcal{H}_2 = N(S) \cap N(T^*)$$

there is an orthogonal decomposition $H_2 = \mathcal{H}_2 \oplus (N(S) \cap N(T^*))^\perp = \mathcal{H}_2 \oplus \overline{R(T)} \oplus \overline{R(S^*)}$, so $N(S) = \mathcal{H}_2 \oplus \overline{R(T)}$

Theorem 7.9 There is $C > 0$ such that for all $y \in D(S) \cap D(T^*)$

$$| y | \leq C(| Sy | + | T^*y |) \tag{7.1}$$

i.e. the *basic estimate* holds⇔ $\mathcal{H}_2 = 0$ and $R(T), R(S^*)$ are closed

Proof. ⇒: If $y \in \mathcal{H}_2$, then $| y | \leq C(| Sy | + | T^*y |) = 0$, hence $\mathcal{H}_2 = 0$. If $y \in R(T) \cap D(T^*)$, then $y \in N(S)$ and $| y | \leq C \, | T^*y |$, by Proposition 7.7, $R(T)$ is closed, similarly, $R(S^*)$ is closed ⇐: $H_2 = R(T) \oplus R(S^*)$ and $y \in D(S) \cap D(T^*)$, write $y = y_1 + y_2$, $y_1 \in R(T) \cap D(T^*)$, $y_2 \in R(S^*) \cap D(S)$. Apply the previous estimates and the triangle inequality

$$| y | \leq | y_1 | + | y_2 | \leq C_1 \, | T^*y | + C_2 \, | Sy | \leq C(| Sy | + | T^*y |)$$

□

$\mathcal{D}_{(p,q)}(\Omega) \subseteq L^2_{(p,q)}(\Omega)$ be the smooth (p,q)-forms with compact support in Ω. Consider the unbounded operator $\bar\partial : L^2_{(p,q)}(\Omega) \to L^2_{(p,q+1)}(\Omega)$ with domain $D(\bar\partial) = \left\{u \in L^2_{(p,q)}(\Omega) \mid \bar\partial u \in L^2_{(p,q+1)}(\Omega)\right\}$, the derivative is in the sense of distributions $\langle \bar\partial u, \alpha\rangle_{L^2} = \langle u, \bar\partial^* \alpha\rangle_{L^2}$ for $\alpha \in \mathcal{D}_{(p,q+1)}(\Omega)$. $\bar\partial^*$ is called the *formal adjoint* of $\bar\partial$. Then $\bar\partial u \in L^2$ if there is a constant $C > 0$ such that $| \langle u, \bar\partial^* \alpha\rangle | \leq C \parallel \alpha \parallel_{L^2}$. In this case, the Hahn-Banach and Riesz representation theorem, $\langle u, \bar\partial^* \alpha\rangle = \langle v, \alpha\rangle$

Proposition 7.10 $\bar\partial : L^2_{(p,q)}(\Omega) \to L^2_{(p,q+1)}(\Omega)$ is a closed operator

Proof. $u_i \to u$ in L^2, $u_i \in D(\bar\partial)$, $\bar\partial u_i \to \alpha$. Let $\beta \in \mathcal{D}_{(p,q+1)}(\Omega)$, then

$$\langle u, \bar\partial^* \beta\rangle = \lim_{i\to\infty} \langle u_i, \bar\partial^* \beta\rangle = \lim_{i\to\infty} \langle \bar\partial u_i, \beta\rangle = \langle \alpha, \beta\rangle$$

So the map $\beta \mapsto \langle u, \bar\partial^* \beta\rangle$ is bounded and $\bar\partial u = \alpha$, thus $\bar\partial$ is closed

□

Example 7.11 Consider $T = i\frac{d}{dx}$ on $L^2([0,1])$, with $D(T)$ consisting of $f, f' \in L^2$. One can show $f \in D(T)$ is (absolutely) continuous on $[0,1]$. By integration by parts

$$f \mapsto \langle Tf, g\rangle = \langle f, Tg\rangle + i(f(1)\bar g(1) - f(0)\bar g(0))$$

This is not continuous with respect to the L^2 topology on f, unless $g(0) = g(1) = 0$. Thus T^* is the same operator T, but with a different domain of definition

The problem is that while compactly supported functions are dense in L^2 topology, they are not dense in the L^2_1 topology, i.e. the graph norm $\| u \| + \| \bar{\partial} u \|$

Methods to fix this

1. Hömander uses a clever choice of (three) weights to prove the basic estimate
2. If Ω has sufficiently nice boundary, define boundary conditions (the $\bar{\partial}$-Neumann problem)
3. Change the geometry of Ω to carry a complete Kähler metric

We will follow the last option (Demially, Gaffney)

Lemma 7.12 (M, g) is a Riemannian manifold, the following are equivalent

1. (M, g) is complete (Hopf-Rinow theorem)
2. $\exists$ compact exhaustion function ψ with $| d\psi |_g \leq 1$
3. $\exists$ compact exhaustion $K_i \subseteq K^\circ_{i+1}$, and $0 \leq \psi_i$ supported in K_{i+1}, $\equiv 1$ on K_i, such that $| d\psi_i |_g \leq 2^{-i}$

Proof. 1.⇒2.: Fix $x_0 \in M$, Let $\psi_0(x) = \frac{1}{2} d(x_0, x)$. Smooth ϕ_0 to ψ with $| \phi - \phi_0 | < 1$, by convolution with some $g \in C^\infty$, compactly supported near 0 and $\int g = 1$ 2.⇒3.: Choose a smooth function $\rho : \mathbb{R} \to [0, 1]$ with $\rho(t) = \begin{cases} 1 & t \leq 1 \\ 0 & t \geq 2 \end{cases}$ and $| \rho'(t) | \leq 2$. Then let $K_i = \left\{ \psi(x) \leq 2^{i+1} \right\}$, $\psi_i(x) = \rho(2^{-i-1} \psi(x))$ 3.⇒2.: Set $\psi = \sum 2^i (1 - \psi_i)$ 2.⇒1.: $| \psi(x) - \psi(y) | \leq d(x, y)$, $\{x_i\}$ is a Cauchy sequence, then $\{x_i\}$ lies in the set $\{\psi \leq C\}$ for some C. Since this is compact, the sequence converges, so (M, g) is complete

□

Corollary 7.13 Let Ω have a complete Riemannian metric ω. Then $\mathcal{D}_{(p,q)}(\Omega)$ is dense in graph norm of $\bar{\partial}$

Proof. Set $u_i = u\phi_i$ as in Lemma 7.12, 3. Then $u_i \to u$ in L^2, and $\bar{\partial} u_i = \bar{\partial} u \psi_i + u \bar{\partial} \psi_i \to \bar{\partial} u$ in L^2. Now choose $v_i \in \mathcal{D}_{(p,q)}(\Omega)$ so that $\| v_i - u_i \|_{L^2_1} = \| v_i - u_i \|_{L^2} + \| \bar{\partial} v_i - \bar{\partial} u_i \|_{L^2} \leq 1/i$, the result follows

□

Corollary 7.14 (Ω, ω) is complete, $\bar{\partial}^*$ with domain

$$D(\bar{\partial}^*) = \left\{ \alpha \in L^2_{(p,q+1)}(\Omega) \;\middle|\; \bar{\partial}^* \alpha \in L^2_{(p,q)}(\Omega) \right\}$$

is the adjoint of T^* of $T = \bar{\partial}$

Proof. $D(T^*) \subseteq D(\bar{\partial}^*)$. Let $u \in \mathcal{D}_{(p,q)}(\Omega) \subseteq D(T)$. If $\alpha \in D(T^*)$, then there is a constant $C > 0$ such that $| \langle \bar{\partial} u, \alpha \rangle | \leq C | u |$. But then by definition, $| \langle u, \bar{\partial}^* \alpha \rangle | \leq C | u |$. Since $\mathcal{D}_{(p,q)}(\Omega)$ is dense in L^2, this implies $\bar{\partial}^* \alpha \in L^2$
$D(\bar{\partial}^*) \subseteq D(T^*)$. If $\bar{\partial}^* \alpha \in L^2$, there is a constant $C > 0$ such that $| \langle u, \bar{\partial}^* \alpha \rangle | \leq C | u |$. Fix $u \in D(T)$. Let $u_i \in \mathcal{D}_{(p,q)}(\Omega)$ so that $u_i \to u$ and $\bar{\partial} u_i \to \bar{\partial} u$ in L^2. Then since $\langle u_i, \bar{\partial}^* \alpha \rangle = \langle \bar{\partial} u_i, \alpha \rangle$, we have $| \langle \bar{\partial} u, \alpha \rangle | \leq C | u |$, and so $\alpha \in D(T^*)$

□

8 Kähler metrics

Definition 8.1 $\Omega \subseteq \mathbb{C}^n$. A *hermitian metric* on Ω is a positive definite hermitian valued smooth function $g = (g_{i\bar{j}})$. The *Kähler form* associated to g is $\omega = i\sum_{i,j=1}^n g_{i\bar{j}} dz_i \wedge d\bar{z}_j$, note that $\bar{\omega} = \omega$. We assume that g is tensorial, in the sense that ω is a well-defined $(1,1)$-form on Ω, $g_{i\bar{j}} = \left\langle \frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j} \right\rangle$. This gives a pointwise hermitian inner product on $(1,0)$-forms: $\alpha = \alpha_i dz_i$, $\beta = \beta_j dz_j$

$$\langle \alpha, \beta \rangle = \sum_{i,j=1}^n \alpha_i \overline{\beta_j} g^{i\bar{j}}$$

$(g^{i\bar{j}})$ is the inverse matrix of (g_{ij}). Extend this to $(p,0)$-forms by

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_p, \beta_1 \wedge \cdots \beta_p \rangle = \det \langle \alpha_i, \beta_j \rangle$$

Extend this to (p,q)-forms by taking the product of this. This gives a complex anti-linear isometry

$$\overline{*} : \Lambda^{p,q} \to \Lambda^{n-p,n-q}, \alpha \wedge \overline{*}\, \beta = \langle \alpha, \beta \rangle \frac{\omega^n}{n!}$$

$\frac{\omega^n}{n!}$ is the volume form. Inducing L^2 inner products

$$\langle \alpha, \beta \rangle = \int_\Omega \alpha \wedge \overline{*}\, \beta$$

Define the *Lefschetz operator*

$$L : \Lambda^{p,q} \to \Lambda^{p+1,q+1}, \alpha \mapsto \omega \wedge \alpha$$

Let $\Lambda = L^*$. Note that $L^n(1) = \omega^n$, $\Lambda^{n,n} \cong \mathbb{C}$

Example 8.2 If F is of type $(1,1)$, write $F = \sum_{i,j} F_{i\bar{j}} dz_i \wedge d\bar{z}_j$. Then $\Lambda F = \sum_{i,j} F_{i\bar{j}} g^{i\bar{j}}$. If α is of type $(1,0)$

$$i\alpha \wedge \bar{\alpha} \wedge \frac{\omega^{n-1}}{(n-1)!} = |\ \alpha\ |^2\ \frac{\omega^n}{n!}$$

Definition 8.3 A hermitian metric is called Kähler if $d\omega = 0$ (ω is closed), equivalently, $\frac{\partial g_{j\bar{k}}}{\partial z_i} = \frac{\partial g_{i\bar{k}}}{\partial z_j}$

Proposition 8.4 ω is Kähler iff about any point there are coordinates such that g is euclidean to order two

Proof. We may always choose local coordinate so that $g_{i\bar{j}}(0) = \delta_{ij}$, $g_{i\bar{j}} = \delta_{ij} + A^k_{i\bar{j}} z_k + B^{\bar{k}}_{i\bar{j}} \bar{z}_k + O(|\ z\ |^2)$. The Kähler condition implies: $A^k_{i\bar{j}} = A^i_{k\bar{j}}$. The fact that $g_{i\bar{j}}$ is hermitian implies $B^{\bar{k}}_{i\bar{j}} = \overline{A^k_{j\bar{i}}}$. Define $w_m = z_m + \frac{1}{2} A^i_{j\bar{m}} z_i z_j$, then $\frac{\partial w_m}{\partial z_j} = \delta_{mi} + A^i_{j\bar{m}} z_i$. Now $\tilde{g}_{m\bar{n}} \frac{\partial w_m}{\partial z_i} \overline{\frac{\partial w_m}{\partial z_i}} = g_{i\bar{j}}$. This implies

$$\begin{aligned} g_{i\bar{j}}(z) &= \delta_{ij} + A^k_{i\bar{j}} z_k + B^{\bar{k}}_{i\bar{j}} \bar{z}_k + O(|\ z\ |^2) \\ &= \tilde{g}_{i\bar{j}} + \tilde{g}_{m\bar{j}} A^k_{i\bar{m}} z_k + \tilde{g}_{i\bar{k}} \overline{A^k_{j\bar{n}} z_k} + O(|\ z\ |^2) \end{aligned}$$

□

Proposition 8.5 (Kähler identities) (Ω, ω) has a Kähler metric. Then the formal L^2 adjoints are given by $\bar{\partial}^* = -i[\Lambda, \partial], \partial^* = i[\Lambda, \bar{\partial}]$

Proof. It suffices to prove these for the euclidean metric. Then is a direct computation

□

Example 8.6 $\Omega \subseteq \mathbb{C}$, $\Lambda(idz \wedge d\bar{z}) = 1$, $f \in \mathcal{D}_{(0,0)}(\Omega)$, $\beta \in \mathcal{D}_{(0,1)}(\Omega)$, $\beta = \beta(z)d\bar{z}$. Then

$$\begin{aligned}
\langle \bar{\partial} f, \beta \rangle &= \int_\Omega \partial_{\bar{z}} f \overline{\beta(z)} i dz \wedge d\bar{z} \\
&= -\int_\Omega f \overline{\partial z \beta(z)} i dz \wedge d\bar{z} \\
&= \int_\Omega f \overline{i \partial \beta} \\
&= \int_\Omega f \overline{\Lambda(i\partial\beta) i dz \wedge d\bar{z}} \\
&= -\int_\Omega f \overline{\Lambda(i\partial\beta)} i dz \wedge d\bar{z} \\
&= \langle f, -i\Lambda\partial\beta \rangle \\
&= \langle f, \bar{\partial}^* \beta \rangle
\end{aligned}$$

9 Solving $\bar{\partial}$ equation

Theorem 9.1 If $\Omega \subseteq \mathbb{C}^n$ is pseudoconnvex, then there is a complete Kähler metric on Ω

Proof. From Richberg's lemma, there is a smooth strictly plurisubharmonic exhaustion function ψ on Ω. By adding a constant, we can assume $\psi > 0$. Let ω_0 denote the euclidean Kähler form on Ω, and consider: $\omega = \omega_0 + i\partial\bar{\partial}\psi^2$, $i\partial\bar{\partial}\psi^2$ is semi-positive definite

$$\omega = \omega_0 + i\partial\psi \wedge \bar{\partial}\psi + i\psi\partial\bar{\partial}\psi \geq \omega_0 + i\partial\psi \wedge \bar{\partial}\psi$$

$$\omega^n \geq \omega_0 \wedge \omega^{n-1} + i\partial\psi \wedge \bar{\partial}\psi \wedge \omega^{n-1} \geq i\partial\psi \wedge \bar{\partial}\psi \wedge \omega^{n-1}$$

$$\frac{\omega^n}{n!} \geq \frac{2}{n} i\partial\psi \wedge \bar{\partial}\psi \wedge \frac{\omega^{n-1}}{(n-1)!} = \frac{2}{n} \mid \partial\psi \mid_\omega^2 \frac{\omega^n}{n!}$$

So $\mid d\psi \mid_\omega = \sqrt{2} \mid \partial\psi \mid_\omega \leq C$. Completeness follows from Lemma 7.12

□

Definition 9.2 Let $d\mu = \omega^n/n!$ be a Kähler metric on Ω, $\phi \in C^0(\Omega)$ be an exhaustion function. Define $L^2_{(p,q)}(\Omega, \phi)$ to be the completion of smooth (p,q)-forms with respect to the norm

$$\| \alpha \|_\phi^2 = \int_\Omega \mid \alpha \mid_\omega^2 e^{-\phi} d\mu$$

The same density theorems apply as for unweighted spaces

For compactly supported α

$$\int_\Omega \langle \bar{\partial} u, \alpha \rangle e^{-\phi} d\mu = \int_\Omega \langle u, e^{\phi} \bar{\partial}^* (e^{-\phi} \alpha) \rangle e^{-\phi} d\mu$$

The new adjoint is $-i[\Lambda, \partial_\phi]$, $\partial_\phi = e^{\phi} \partial e^{-\phi} = \partial - \partial\phi$. Moreover

$$\int_\Omega \langle \partial_\phi u, \alpha \rangle e^{-\phi} d\mu = \int_\Omega \langle \partial (e^{-\phi} u), \alpha \rangle d\mu = \int_\Omega \langle u, \bar{\partial}^* \alpha \rangle e^{-\phi} d\mu$$

So $\partial_\phi^* = i[\Lambda, \bar{\partial}]$

Definition 9.3 The Laplacian is $\Delta = d^* d + d d^*$. The Dolbeault laplacians are

$$\Box_{\bar{\partial}} = \bar{\partial}^* \bar{\partial} + \bar{\partial}\bar{\partial}^*, \Box_\partial = \partial_\phi^* \partial_\phi + \partial_\phi \partial_\phi^*$$

The curvature is the pure imaginary $(1,1)$ form

$$F_\phi = \bar{\partial}\partial_\phi + \partial_\phi \bar{\partial} = \partial\bar{\partial}\phi$$

Lemma 9.4 $\Box_{\bar{\partial}} - \Box_\partial = [iF_\phi, \Lambda]$

Proof.

$$\bar{\partial}^* \bar{\partial} = -i[\Lambda, \partial_\phi]\bar{\partial} = -i\Lambda\partial_\phi\bar{\partial} + i\partial_\phi\Lambda\bar{\partial}$$

$$\bar{\partial}\bar{\partial}^* = \bar{\partial}(-i[\Lambda, \partial_\phi]) = -i\bar{\partial}\Lambda\partial_\phi + i\bar{\partial}\partial_\phi\Lambda$$

$$-\partial_\phi^* \partial_\phi = -i[\Lambda, \bar{\partial}]\partial_\phi = -i\Lambda\bar{\partial}\partial_\phi + i\bar{\partial}\Lambda\partial_\phi$$

$$-\partial_\phi \partial_\phi^* = -\partial_\phi(i[\Lambda, \bar{\partial}]) = -i\partial_\phi\Lambda\bar{\partial} + i\partial_\phi\bar{\partial}\Lambda$$

□

Corollary 9.5 For $\alpha \in D(\bar{\partial}) \cap D(\bar{\partial}^*)$

$$\| \bar{\partial}\alpha \|^2 + \| \bar{\partial}^*\alpha \|^2 \geq \langle [iF_\phi, \Lambda]\alpha, \alpha \rangle$$

Proof. $\langle \partial_\phi \alpha, \alpha \rangle \geq 0$ can be throw away, and $\langle \Box_{\bar{\partial}} \alpha, \alpha \rangle$ give the left hand side by integration by parts

□

Lemma 9.6 Write $iF_\phi = f_{i\bar{j}} dz_i \wedge d\bar{z}_j$. If there is $C_0 > 0$ such that $\sum f_{i\bar{j}}(z)\xi_i\bar{\xi}_j \geq C_0 \mid \xi \mid^2$ for all $\xi \in \mathbb{C}^n$ and all $z \in \Omega$, then there is $C_1 > 0$ such that

$$\langle [iF_\phi, \Lambda]\alpha, \alpha \rangle \geq C_1 \| \alpha \|^2$$

for all $\alpha \in L^2_{(n,q)}(\Omega, \omega), q \geq 1$

Note that $[iF_\phi, \Lambda] = 0$ on $(n, 0)$ forms. In particular, the condition that $q \geq 1$ is necessary. The applications of theis result extends to (p, q) froms, $q \geq 1$. We will only prove a couple of special cases. For the general result see Demailly

Example 9.7 Consider an $(n, 1)$ form α. Let θ_i denote an orthonormal frame for $T^{1,0}\Omega$. Write $\omega = i\sum_j \theta_j \wedge \bar{\theta}_j$, and

$$\alpha = i\sum_{i=1}^n \alpha_i(z)\theta_1 \wedge \cdots \wedge \theta_n \wedge \bar{\theta}_i$$

Write $iF_\phi = \sum f_{i\bar{j}}\theta_i \wedge \bar{\theta}_j$

Write $\sum_{i,j} \frac{\partial^2 \phi_0}{\partial z_i \partial \bar{z}_j}(z)\xi_i\bar{\xi}_j \geq m(z) \mid \xi \mid^2$, where m is a continuous function, $m(z) > 0$ for all $z \in \Omega$. We will replace a given ϕ_0 with $\phi = \chi \circ \phi_0$, for an appropriate increasing convex function χ

Set $M(t) = (\min_{\phi_0(z) \geq t} m(z))^{-1}$. Then $M(t)$ is a positive, continuous, increasing function of t. Note that $M(\phi_0(z))m(z) \geq 1$

Claim. We can find a smooth $\widetilde{M} \geq M$ that is increasing

Assuming the claim, set $\chi(t) = \int^t \widetilde{M}(\tau)d\tau$. Then χ is convex, and $\chi'(t) \geq M(t)$. Set $\phi = \chi \circ \phi_0$. Then ϕ is psh exhaustion function, and

$$\sum_{i,j} \frac{\partial^2 \phi}{\partial z_i \partial \bar{z}_j}(z)\xi_i\bar{\xi}_j \geq \sum_{i,j} \chi' \circ \phi_0 \frac{\partial^2 \phi_0}{\partial z_i \partial \bar{z}_j}(z)\xi_i\bar{\xi}_j \geq M(\phi_0(z))m(z) \mid \xi \mid^2 \geq \mid \xi \mid^2$$

Proof of claim. This is probably obvious, by here is one idea: Let $\psi : \mathbb{R} \to \mathbb{R}$ be smooth with compact support in $(-1, 1)$, $0 \leq \psi \leq 1$ and $\int_{\mathbb{R}} \psi = 1$. Set

$$\widetilde{M}(t) = \int_{\mathbb{R}} M(s)\psi(s - t - 1)ds = \int_{\mathbb{R}} M(\tau + t + 1)\psi(\tau)d\tau$$

The first equality proves that $\widetilde{M}$ is smooth. By the second equality

$$\widetilde{M}(t) \geq \int_{\mathbb{R}} M(t)\psi(\tau)d\tau = M(t)$$

Also by the second equality, for $\delta > 0$

$$\widetilde{M}(t + \delta) - M(t) = \int_{\mathbb{R}} (M(\tau + t + 1) - M(\tau + t + 1))\psi(\tau)d\tau \geq 0$$

So $\widetilde{M}$ is increasing

□

Summary: $\Omega \subseteq \mathbb{C}^n$ pseudoconvex, ω complete. For an (n,q) form α, $q \geq 1$, and $\alpha \in D(\bar{\partial}) \cap D(\bar{\partial}^*)$, we have proved the basic estimate

$$\| \bar{\partial}\alpha \|_{\phi} + \| \bar{\partial}^*\alpha \|_{\phi} \geq C \| \alpha \|_{\phi}$$

This implies that $\bar{\partial}$ has closed range and that the harmonics $\mathcal{H}^{n,q} = \{0\}$, i.e. $\ker \bar{\partial} = \text{im } \bar{\partial}$

Theorem 9.8 If $\alpha \in L^2_{(n,q)}(\Omega, \phi)$, $q \geq 1$, and $\bar{\partial}\alpha = 0$, then there is $u \in L^2_{(n,q-1)}(\Omega, \phi)$ such that $\bar{\partial}u = \alpha$. Moreover, $\| u \| \leq C \| \alpha \|$

Note. In applications, having the estimate on the norms is crucial. Notice that $\|\|$ depends on the Kähler metric ω and the weight ϕ. We can get rid of the dependence on ω, i.e. prove the same result for the euclidean metric, by taking a limit of solutions for metrics $\omega_\epsilon = \omega_0 + \epsilon\omega$ as $\epsilon \to 0$ (see Demailly). On the other hand, ϕ is necessary. Optimizing the choice of ϕ is an important issue we will skip for now. Finally, the result holds more generally for (p,q) forms: $\bigwedge^p T^*\Omega \cong \bigwedge^{n-p} T\Omega \otimes \bigwedge T^*\Omega$, so trivializing $\bigwedge^{n-p} T\Omega$ reduces the problem to this case. More importantly. This is not exactly what we want, since we are trying to solve $\bar{\partial}$ for all smooth α, not just those that are integrable. Let $L^2_{(p,q)}(\Omega, \text{loc})$ denote the (p,q) forms that are locally L^2. Notice that here the metric ω and weight ϕ are irrelevant

Theorem 9.9 If $\alpha \in L^2_{(p,q)}(\Omega, \text{loc})$, $q \geq 1$, and $\bar{\partial}\alpha = 0$, then there is $u \in L^2_{(p,q-1)}(\Omega, \text{loc})$ such that $\bar{\partial}u = \alpha$

Proof. The idea is to find some weight $\tilde{\phi}$ such that $\alpha \in L^2_{(p,q)}(\Omega, \tilde{\phi})$, and then apply the previous result. We again choose the form $\tilde{\phi} = \chi \circ \phi$, where χ is increasing and convex. To do this, let $K_i = \{z \in \Omega \mid \phi(z) \leq i\}$, and let $\ell_i = \| \alpha \|^2_{L^2(K_i)}$. Now choose χ so that $e^{-\chi(i)}(\ell_{i+1} - \ell_i) \leq 2^{-i}$, then

$$\int_{K_{i+1}\setminus K_i} | \alpha |^2 \, e^{-\tilde{\phi}} d\mu \leq e^{-\chi(i)} \int_{K_{i+1}\setminus K_i} | \alpha |^2 \, d\mu \leq e^{-\chi(i)}(\ell_{i+1} - \ell_i) \leq 2^{-i}$$

Hence

$$\int_{\Omega} | \alpha |^2 \, e^{-\tilde{\phi}} d\mu = \sum_i \int_{K_{i+1}\setminus K_i} | \alpha |^2 \, e^{-\tilde{\phi}} d\mu \leq \sum_i 2^{-i} < \infty$$

□

Elliptic regularity: The result applies, in particular, to the case where α is smooth. However, the theorem just concludes that the solution $\bar{\partial}u = \alpha$ is locally in L^2. We want to improve this

Theorem 9.10 If α is a smooth (p,q)-form on Ω, $q \geq 1$, with $\bar{\partial}\alpha = 0$, then there is a smooth $(p, q-1)$-form on Ω such that $\bar{\partial}u = \alpha$

Proof. First consider the case $q = 1$, i.e. u is a function. We know that $\partial^*\partial = \bar{\partial}^*\bar{\partial}$, so $\| \partial u \| \leq \| \bar{\partial}u \|$. In particular, if $\alpha \in L^2(\Omega, \text{loc})$, then u is in $L^2_1(\Omega, \text{loc})$ (one distributional derivative in L^2). Now differentiate the equation $\bar{\partial}u = \alpha$ k times to conclude that $u \in L^2_k(\Omega, \text{loc})$ for arbitrary k. On the other hand, the Sobolev embedding theorem implies $L^2_k \hookrightarrow C^j$ for $k \geq n + j$. Hence, if α is smooth, so is u
If $q \geq 2$, then we have

$$L^2_{(p,q-1)}(\Omega, \phi) = R(\bar{\partial}) \oplus R(\bar{\partial}^*)$$

Moreover, $N(\bar{\partial})^\perp = R(\bar{\partial}^*)$. Hence, u may be taken to be in the range of $\bar{\partial}^*$. Since $(\bar{\partial}^*)^2 = 0$, we have $\Box_{\bar{\partial}} u = \bar{\partial}^*\bar{\partial}u = \bar{\partial}^*\alpha$. Let $\Delta = dd^* + d^*d$ be the ordinary or de Rham Laplacian. Then in standard basis and euclidean metric, Δ acts on the coefficients of (p,q)-forms

□

Claim. $\Delta = 2\Box_{\bar\partial}$

Proof. Write $d = \partial + \bar\partial$, $d^* = \partial^* + \bar\partial^*$, then

$$\Delta = \Box_{\bar\partial} + \Box_{\partial} + \bar\partial\partial^* + \partial^*\bar\partial + \partial\bar\partial^* + \bar\partial^*\partial$$

But the cross terms vanish: e.g. since $\bar\partial^2 = 0$

$$\bar\partial\partial^* = i\bar\partial[\Lambda, \bar\partial] = i\bar\partial\Lambda\bar\partial$$

$$\partial^*\bar\partial = i[\Lambda, \bar\partial]\bar\partial = -i\bar\partial\Lambda\bar\partial$$

The result now follows, since $\Box_\partial = \Box_{\bar\partial}$

□

It follows that if $\alpha \in L^2_1(\Omega, \mathrm{loc})$, then $\Delta u = 2\bar\partial^*\alpha$ is in $L^2(\Omega, \mathrm{loc})$. We now appeal to the following interior estimate: if $U \subset\subset U' \subset\subset \Omega$, then there is a constant $C > 0$ such that

$$\| u \|_{L^2_{k+1}(U)} \leq C\left(\| \|_{L^2_k(U')} + \| \Delta u \|_{L^2_k(U')} \right)$$

for all smooth (p, q)-forms. By "bootstrapping" the equation, we conclude that if $\bar\partial u = \alpha$, $\bar\partial^* u = 0$, for α smooth, then u is in $L^2_k(\Omega, \mathrm{loc})$ for any k, and hence is smooth by Sobolev embedding

Theorem 9.11 If $\Omega \subseteq \mathbb{C}^n$ is pseudoconvex, then $H^k_{\mathrm{dR}}(\Omega) = 0$ for $k > n$

Remark 9.12 This is sharp, i.e. it is possible that $H^n_{\mathrm{dR}}(\Omega) = 0$. For example $A_i = \left\{ \frac{1}{2} < \mid z_i \mid < 2 \right\}$, $\Omega = A_1 \times \cdots \times A_n$, let $Y = S^1 \times \cdots \times S^1 \subset \Omega$, and

$$\beta = \frac{dz_1 \wedge \cdots \wedge dz_n}{z_1 \cdots z_n}$$

Then $d\beta = \bar\partial\beta = 0$, and

$$\int_Y \beta = \int_{\mid z_1 \mid = 1} \frac{dz_1}{z_1} \cdots \int_{\mid z_n \mid = 1} \frac{dz_n}{z_n} = (2\pi i)^n$$

Hence $0 \neq [\beta] \in H^n_{\mathrm{dR}}(\Omega)$

Claim. Suppose β is a k-form such that $d\beta$ is of type $(k+1, 0)$. Then β is cohomologous to a $(k, 0)$-form

Proof. Write $\beta = \sum_{q=0}^{l} \beta_{(k-q,q)}$. Proof by induction on l. The case $l = 0$ is trivial. Assume $l \geq 1$ and that the result holds for $l - 1$. We have

$$d\beta = \partial\beta + \bar\partial\beta = \sum_{q=0}^{l} \partial\beta_{(k-q,q)} + \bar\partial\beta_{(k-q,q)}$$

Thus $\bar\partial\beta_{(k-l,l)} = 0$, $\beta_{(k-l,l)} = \bar\partial u$, for a smooth $(k-l, l-1)$ form u. Let $\tilde\beta = \beta - du$. Then $\tilde\beta = \sum_{q=0}^{l-1} \tilde\beta_{(k-q,q)}$, $d\beta = d\tilde\beta$,

□

Remark 9.13 Another proof use Morse theory: ϕ is a C^∞ psh exhaustion function, Morse function on Ω. The complex Hessian $\frac{\partial^2}{\partial z_j \partial \bar z_j}\phi > 0$ bound the number of negative eigenvalues of the real Hessian
Lefschetz hyperplane theorem:

Solution to the Levi problem

Theorem 9.14 $\Omega \subseteq \mathbb{C}^n$ is a domain of holomorphy iff it is pseudoconvex. We have already proved $\Rightarrow$, by showing that $-\log d(z)$ is a psh exhaustion function. We also showed that being a domain of holomorphy is equivalent to being holomorphically convex

Theorem 9.15 If we can solve $\bar{\partial}$ on $\Omega \subseteq \mathbb{C}^n$, then it is a domain of holomorphy

Proof. Let $\tilde{\alpha} \in C^\infty_{(p,q)}(\tilde{\Omega})$, $q \geq 1$, and $\bar{\partial}\tilde{\alpha} = 0$. Set $\beta = z_n^{-1}\bar{\partial}\phi \wedge \pi^*\tilde{\alpha}$. Then $\beta \in C^\infty_{(p,q+1)}(\Omega)$. Moreover, $\bar{\partial}\beta = 0$. By the hypothesis of the theorem, there is $u \in C^\infty_{(p,q)}(\Omega)$ such that $\bar{\partial}u = \beta$. Consider $\alpha = \phi\pi^*\tilde{\alpha} - z_n u$. Then $\bar{\partial}\alpha = 0$. Hence, there is $v \in C^\infty_{(p,q-1)}(\Omega)$ such that $\bar{\partial}v = \alpha$. Finally, since z_n vanishes and $\phi \equiv 1$ on $j(\tilde{\Omega})$ This proves the claim. By the induction hypothesis, $\tilde{\Omega}$ is a domain of holomorphy. Hence, there is $\tilde{f} \in A(\tilde{\Omega})$ that blows up at ξ. As in the first part of the argument above, there is $u \in C^\infty(\Omega)$ such that $f = \phi\pi^*\tilde{f} - z_n u \in A(\Omega)$, and $j^*f = \tilde{f}$. Hence, f blows up at ξ

□

10 Proper mapping theorems

Theorem 10.1 (Remmert theorem) If $f : X \to Y$ is proper and $V \subseteq X$ is an analytic subvariety of X, then $f(V) \subseteq Y$ is an analytic subvariety of Y

Theorem 10.2 (Remmert-Stein theorem) $V \subseteq X$ is an analytic subvariety, $W \subseteq X \setminus V$ is an irreducible analytic subvariety. If $\dim V < \dim W$, then the $\overline{W} \subseteq X$ is an irreducible analytic subvariety

Remark 10.3 The assumption on dimension is necessary. For example, take the graph in $\mathbb{C}^2 \subseteq \mathbb{CP}^2$ of an entire function f. Suppose f has essential singularity at infinity

Corollary 10.4 X is an analytic space, V is an analytic subvariety, $W \subseteq X \setminus V$ is an analytic subvariety. If $\dim V$ is less than the dimension of any irreducible components of W, then $\overline{W} \subseteq X$ is an analytic subvariety

Theorem 10.5 (Chow's theorem) $V \subseteq \mathbb{CP}^n$ is an analytic subvariety, then V is a projective algebraic variety

Remark 10.6 X is a Kahler manifold. X is a Moisezon manifold(meaning the algebraic of dimension of $X = \dim X$). A theorem of Moisezon says Kahler + Moisezon $\Rightarrow$ projective. There is an equivalence between Hodge manifolds(Kahler manifolds with integral Kahler class) and projective algberaic manifolds

Corollary 10.7 Every compact Riemann surface is an algebraic curve A holomorphic map between nonsingular projective algebraic varieties is a morphism of varieties Every memomorphic function on $\mathbb{CP}^n$ is rational

Definition 10.8 $\phi : X \to Y$ is holomorphic. $\mathcal{E} \to X$, $\mathcal{F} \to Y$ are sheaves of $\mathcal{O}_X$, $\mathcal{O}_Y$ modules. The sheaf $\phi^{-1}\mathcal{F}$ has presheaf is intuitively $\phi^{-1}\mathcal{F}(U) = \mathcal{F}(\phi(U))$. Alternatively, the espace étalé $X \times_Y \mathcal{F}$, at the level of stalks, $(\phi^{-1}\mathcal{F})_p = \mathcal{F}_{\phi(p)}$. By composition with ϕ defines a sheaf map $\phi^{-1}\mathcal{O}_Y \to \mathcal{O}_X$, making $\mathcal{O}_X, \mathcal{E}$ into $\phi^{-1}\mathcal{O}_Y$ modules

$$\phi^*\mathcal{F} = \mathcal{O}_X \otimes_{\phi^{-1}\mathcal{O}_Y} \phi^{-1}\mathcal{F}$$

There are canonical maps $\phi^*(\phi_*\mathcal{E}) \to \mathcal{E}$, $\mathcal{F} \to \phi_*(\phi^*\mathcal{F})$ through

$$\mathcal{E}(\phi(\phi^{-1}(U))) \to \mathcal{E}(U), \mathcal{F}(U) \to \mathcal{F}(\phi^{-1}(\phi(U)))$$

More generally, there is a canonical bijection $\mathrm{Hom}_{\mathcal{O}_X}(\phi^*\mathcal{F}, \mathcal{E}) \leftrightarrow \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{F}, \phi_*\mathcal{E})$

Fact 10.9 ϕ_* is left exact. ϕ^* is right exact. $\phi^*\mathcal{O}_Y = \mathcal{O}_X$. If $\mathcal{F}$ is coherent, then $\phi^*\mathcal{F}$ is coherent. $\phi_*\mathcal{E}$ is not necessarily coherent even if $\mathcal{E}$ is. For example, let $\phi : \mathbb{C} \to \mathbb{C}^*$ be the exponential map, then $\phi_*\mathcal{O}_\mathbb{C}$ is infinitely generated

Theorem 10.10 (Grauert-Remmert, Direct image theorem) $\phi : X \to Y$ is finite, $\mathcal{E} \to X$ is coherent, then $\phi_*\mathcal{E}$ is coherent. Moreover, ϕ_* is right exact

Example 10.11 $X = \mathbb{C}^2 \setminus \{0\}$, $Y = \mathbb{C}$, $\phi(z_1, z_2) = z_1$. Let $D = \{(0, z_2) \in X\} \cong \mathbb{C}^*$. $\phi(D) = \{0\}$, let $\mathcal{I}_D$ be the ideal sheaf, $\mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}_D \to 0$, let $f = 1/z_2$ is holomorphic on D, but is not in the image of $\phi_*\mathcal{O}_X \to \phi_*(\mathcal{O}_X/\mathcal{I}_D)$ by Hartogs's theorem. Thus ϕ_* is not right exact

Example 10.12 $X = \{0\}$, $Y = \mathbb{C}$, ϕ is the inclusion. Consider injection $0 \to \mathcal{O}_Y \xrightarrow{\times z} \mathcal{O}_Y$. The induced map $\phi^*\mathcal{O}_Y \to \phi^*\mathcal{O}_Y$ is zero, hence ϕ^* is not left exact

$R^i\phi_*\mathcal{E}(V) = H^i(\phi^{-1}(V), \mathcal{E})$

Problem: $\mathcal{E} \to X$ is coherent, $R^i\phi_*\mathcal{E} \to Y$ might not be coherent

Theorem 10.13 (Grauert's proper coherence theorem) X, Y are complex spaces. $\phi : X \to Y$ is proper, $\mathcal{E} \to X$ is coherent, then $R^i\phi_*\mathcal{E} \to Y$ is coherent for all i

Note. If Y is a point, then X is compact, this gives the Cartan-Serre theorem

11 Stein manifolds

Definition 11.1 Let M be a manifold. An almost complex structure on M is an endomorphism $J : TM \to TM$ with $J^2 = -I$. This imipliesd that M has even real dimension $2n$. Let $T^{1,0}M \oplus T^{0,1}M = T_{\mathbb{C}}M = TM \otimes \mathbb{C}$ be the $\pm i$ eigenbundles of J. There is a real isomorphism $\rho : TM \to T^{1,0}M$ such that $\rho J = i\rho$. We say that J is integrable if $T^{1,0}M$ is involutive (Frobenius theorem)

$$[T^{1,0}M, T^{1,0}M] \subseteq T^{1,0}M$$

Let J_0 denote the standard almost complex structure on $\mathbb{R}^{2n}$. Then J is integrable if about every pint there is a coordinate chart (U, ϕ), $\phi : U \to \mathbb{R}^{2n}$ such that $\phi_* J = J_0 \phi_*$. In this case, a local frame for $T^{1,0}M$ on U is given by the complex vector fields $\frac{\partial}{\partial z_1}, \cdots, \frac{\partial}{\partial z_n}$

Definition 11.2 A $2n$ dimensional manifold M is a complex manifold if it admits an atlas $\{(U_\alpha, \phi_\alpha)\}$ where the transition functions $\phi_\beta \circ \phi_\alpha^{-1}$ are biholomorphisms

Note. A manifold may admit inequivalent complex structures. A complex manifold has an integrable associated almost complex structure by pulling back J_0

Theorem 11.3 (Newlander-Nirenberg) An almost complex manifold is integrable iff it admits a complex manifold structure

Definition 11.4 A Hermitian metric on E is a smooth Hermitian inner product on the fiber of $E \to M$. We denote this h or $\langle , \rangle$. If $\{s_i\}_{i=1}^r$ is a holomorphic frame, we write $h_{i\bar{j}} = \langle s_i, s_j \rangle$. A connection on E is a linear map $d_A : \Omega^0(M, E) \to \Omega^1(M, E)$ satisfying the Leibniz rule: $d_A(fs) = df \otimes s + f d_A s$. We say d_A is compatible with a holomorphic structure on E if $\bar{\partial}_A = (d_A)^{0,1} = \bar{\partial}_E$. We say d_A is unitary if for any sections s_1, s_2

$$d\langle s_1, s_2 \rangle = \langle d_A s_1, s_2 \rangle + \langle s_1, d_A s_2 \rangle$$

Proposition 11.5 A Hermitian, holomorphic bundle admits a unique compatible, unitary connection. This is called the Chern connection

Proof. $d_A = \partial_A + \bar{\partial}_E$ is determined by the equation

$$\partial \langle s_1, s_2 \rangle = \langle \partial_A s_1, s_2 \rangle + \langle s_1, \bar{\partial}_A s_2 \rangle$$

In a holomorphic frame s_i, $\partial_A s_i = d_A s_i = \omega_i^j s_j$, where $\omega_i^j = h^{j\bar{k}} \frac{\partial h_{j\bar{k}}}{\partial z_j}$. Or, $\omega = h^{-1} \partial h$

□

Definition 11.6 The curvature of a connection A is $F_A = d_A^2$. This is an endomorphism-valued 2-form, i.e. $F_A \in \Omega^2(M, \text{End } E)$. $\bar{\partial}_E$ is integrable iff $\bar{\partial}_A^2 = F_A^{0,2} = 0$. If A is in addition unitary, then we have that F_A is of type $(1,1)$. In holomorphic local frame, $F_A = \bar{\partial}(h^{-1}\partial h)$. In particular, if E is a line bundle, then $F_A = \bar{\partial}\partial \log h$. In general, if we modify the metric on E by $e^{-\phi}$, then $\omega_\phi = \omega - \partial\phi$ and $F_{(\bar{\partial}_E, he^{-\phi})} = F_{(\bar{\partial}_E, h)} + \partial\bar{\partial}\phi$

Definition 11.7 $(\bar{\partial}_E, h)$ is (strictly) Nakano positive if the Hermitian form $iF_{(\bar{\partial}_E, h)}$ is positive definite, i.e. there is $C > 0$ such that for each $\xi \in (TX)^{1,0} \otimes E$

$$\sum_{i,j=1}^n \langle F_{i\bar{j}} \xi^i, \xi^j \rangle \geq C \mid \xi \mid^2$$

Where

$$F_{(\bar{\partial}_E, h)} = \sum F_{i\bar{j}} dz_i \wedge d\bar{z}_j$$

A holomorphic line bundle $\mathcal{L} \to X$ is positie if it admits a Nakano positive Hermitian metric. If ϕ is smooth and strictly plurisubharmonic, then by replacing ϕ by $\chi \circ \phi$ for a suitable positive convex function, we can arrange for $F_{(\bar{\partial}_E, he^{-\phi})}$ to be Nakano positive

Theorem 11.8 X is a complex manifold, and there is a smooth strictly plurisubharmonic exhaustion function on X. Let $\mathcal{E} \to X$ be a holomorphic vector bundle. Then for every $\alpha \in \Omega^{p,q}(X,E), q \geq 1$ with $\bar{\partial}_E \alpha = 0$, there is $u \in \Omega^{p,q-1}(X,E)$ such that $\bar{\partial}_E u = \alpha$

If ϕ is the psh function, then $\omega = i\partial\bar{\partial}\phi^2$ is a complete Kähler metric on X. The Kähler identities are still valid for the Chern connection

$$\bar{\partial}_A^* - i[\Lambda, \partial_A], \partial_A^* = i[\Lambda, \bar{\partial}_A]$$

Theorem 11.9 (Holomorphic approximation) $\mathcal{E} \to X$ as above, $K_0 = \{z \in M \mid \phi(z) \leq 0\}$. Then any holomorphic section of $\mathcal{E}$ defined in a neighborhood of K_0 can be uniformly approximated by elements of $H^0(M, \mathcal{E})$

Proof. Let $u \in H^0(U, \mathcal{E})$ for $K_0 \subseteq U$. For any $\epsilon > 0$ we need to find $\tilde{u} \in H^0(X, \mathcal{E})$ such that $\| \tilde{u} - u \|_{K_0} < \epsilon$

Claim. It suffices to show that if $v \in L^2(K,E)$ such that $\langle u, v \rangle_{L^2(K)} = 0$ for all $u \in H^0(X, \mathcal{E})$, then $\langle u, v \rangle_{L^2(K)} = 0$ for all $u \in H^0(U, \mathcal{E})$

Proof of claim. Let $r : H^0(X, \mathcal{E}) \to L^2(K,E)$ be the restriction map. Suppose $u \in H^0(U, \mathcal{E})$ cannot be approximated on K. Define a linear map

$$T : \text{Span } \{\text{im } r, u\} \subseteq L^2(K,E) \to \mathbb{C}$$

$$T(u) = 1, T \mid_{\text{im } r} = 0 \text{ and extending linearly}$$

Then T is continuous, and so by the Hahn-Banach theorem T extends to a bounded operator on $L^2(K,E)$. But then by the Riesz representation theorem there is $v \in L^2(K,E)$ such that $T(\cdot) = \langle \cdot, v \rangle$. This would contradict the conclusion of the claim. By composing with a convex function we find smooth psh functions ϕ_m such that $\phi_m = \phi$ on K_0 and $\phi_m \to +\infty$ on $X \setminus K_0$. By the closed range property (with weight ϕ_m):

$$L^2(X, E, \phi_m) = H^0(X, \mathcal{E}) \oplus R((\bar{\partial}_E)^{*m})$$

Extend ve^{ϕ} by zero to all of X. By the main theorem we can solbe $(\bar{\partial}_E)^{*m} \alpha_m = ve^{\phi}$. Moreover, we have estimates

$$\| \alpha_m \|_{L^2(\phi_m)} \leq C \| ve^{\phi} \|_{L^2(\phi_m)} \leq C \| ve^{\phi} \|_{L^2(\phi)}$$

Let $\beta_m = e^{-(\phi_m - \phi)} \alpha_m$. Then $(\bar{\partial}_E)^* \beta_m = ve^{\phi}$, and

$$\| \beta_m \|_{L^2(\phi)} \leq \| \alpha_m \|_{L^2(\phi_m)} \leq \| ve^{\phi} \|_{L^2(\phi)}$$

Let $\beta_m \to \beta$ weakly in $L^2(\phi)$. Then $\beta \equiv 0$ outside K_0. Use

$$| \beta_m |^2 e^{-\phi} = e^{\phi_m - \phi} | \alpha_m |^2 e^{-\phi_m}$$

Moreover, $(\bar{\partial}_E)^* \beta = ve^{\phi}$ in the sense of distributions. Now by integration by parts: if $u \in H^0(U, \mathcal{E})$, then

$$\begin{aligned} \langle u, v \rangle_{L^2(K_0)} &= \langle u, (\bar{\partial}_E)^* \beta \rangle_{L^2(K_0, \phi)} \\ &= \langle u, (\bar{\partial}_E)^* \beta \rangle_{L^2(U, \phi)} \\ &= \langle \bar{\partial}_E u, \beta \rangle_{L^2(U, \phi)} \\ &= 0 \end{aligned}$$

□

□

Definition 11.10 An n-dimensional complex manifold X is a *Stein manifold* if

1. X is holomorphically convex
2. $A(X)$ separates points, i.e. for $z_1 \neq z_2 \in X$, $\exists f \in A(X)$ such that $f(z_1) \neq f(z_2)$
3. For each $z_0 \in X$ there are $f_1, \cdots, f_n \in A(X)$ that give a coordinate system in a neighborhood of z, i.e. $d_{z_0} f_1, \cdots, d_{z_0} f_n$ are linearly independent

Note. This is vacuous if X is a domain in $\mathbb{C}^n$, just take coordinate functions

Theorem 11.11 X is a Stein iff there exists a smooth strictly psh exhaustion function on X

Proof.

□

Example 11.12

1. A domain of holomorphy in $\mathbb{C}^n$ is Stein
2. A properly embedded complex submanifold is Stein iff it can be properly embedded in a Stein manifold
3. A positive dimensional Stein manifold is necessarily noncompact
4. If $X = 1$ (Riemann surfaces), then X is Stein iff it is noncompact
5. Let X be a compact and $\mathcal{L} \to X$ a positive line bundle. Suppose $0 \not\equiv s \in H^0(X, \mathcal{L})$ with zero set $Z(s)$. Then $X_s = X \setminus Z(s)$ is Stein

Proof.

1. This will follow from the existence of a strongly psh exhaustion function
2. Follows by considering $k \times k$ minors of DF
3. This follows because holomorphic functions on compact (connected) X are constant by the maximum principle
4. ($\Leftarrow$) is a theorem of Behnke-Stein
5. $\phi = -\log| \ s \ |^2$ is a psh exhaustion function on X_s

□

Proposition 11.13 Let X be a Stein manifold, and $\hat{K} \subseteq U$. Then there is a smooth psh exhaustion function ϕ negative on K and positive on U^c

Proof.

□

Lemma 11.14 Suppose X admits a stricly psh exhaustion function ϕ, given $z_0 \in X$ there is U_0 and $u_0 \in A(U_0)$ such that $u_0(z_0) = 0$ and $\text{Re } u_0(z) < \phi(z) - \phi(z_0)$ for $z \neq z_0$ in U_0

Proof. Fix local coordinates where $z_0 = 0$. By Taylor expansion

$$\phi(z) = \phi(0) + \text{Re } P(z) + \sum —(0) z_i \bar{z}_j + O(| \ z \ |^3)$$

Where $P(z)$ is a polynomial of degree ≤ 2, $P(0) = 0$. Now use the fact that teh hessian term is bounded below by $C \mid z \mid^2$

□

Theorem 11.15 Let X be an n-dimensional Stein manifold. Then there is a proper embedding of X in $\mathbb{C}^{2n+1}$

12 Oka's theorem

Cousin problems: Let X be a complex manifold, $\{U_\alpha\}$ a locally finite open covering of X
I. Given $\{m_\alpha\}$ locally meromorphic functions such that $m_\alpha - m_\beta \in A(U_\alpha \cap U_\beta)$, find a global meromorphic function m on X such that $m - m_\alpha$ is holomorphic on each U_α
II. Given $\{m_\alpha\}$ locally meromorphic functions, $m_\alpha \not\equiv 0$, such that $m_\alpha m_\beta^{-1} \in A(U_\alpha \cap U_\beta)$, find a global meromorphic function m on X such that $m m_\alpha^{-1}$ and $m^{-1} m_\alpha$ are holomorphic on each U_α
These generalize the Mittag-Leffler and Weierstrass theorems in one dimension

Solutio to Cousin I for Stein manifold Set $u_{\alpha\beta} = m_\alpha - m_\beta$. Then $\left\{u_{\alpha\beta}\right\}$ is a 1-cocucle of holoomorphic funtions. It suffices to find $u_\alpha \in A(U_\alpha)$ such that $u_{\alpha\beta} = u_\alpha - u_\beta$. For then $m_\alpha - u_\alpha$ is a well-defined global meromorphic funtion. Let $\{\psi_\alpha\}$ be a partition of unity. Set $h_\alpha = \sum_\beta u_{\alpha\beta}\psi_\beta$. Then $u_{\alpha\beta} = h_\alpha - h_\beta$. $\left\{\bar\partial h_\alpha\right\}$ defines a global $(0,1)$-form on X. Since X is Stein, we can find a function v so that $\bar\partial v = \bar\partial h_\alpha$, i.e. $v - h_\alpha$ is holomorphic on each U_α. Set $u_\alpha = h_\alpha - v$

Definition 12.1 Given $\xi \in \mathbb{C}^n$, let $\mathcal{O}_\xi$ denote the ring of germs of holomorhic functions at ξ. $\mathcal{O}_\xi$ is a commutative algebra over $\mathbb{C}$ and $\cong$ the ring of convergent power series at ξ. Clearly, $\mathcal{O}_0^{(n)}$ is an integral domain. Its function field $\mathcal{M}_0$ is the field of germs of meromorphic functions at 0. $\mathcal{O}_0$ is a local ring with residue field $\mathbb{C}$, the unique maximal ideal $\mathfrak{m}_0$ consists of germs of functions vanishing at 0

Definition 12.2 $p \in O_0^{(n-1)}[z_n]$ is called a Weierstass polynomial if it is representaed by a funciton ofthe form

$$p(w, z_n) = z_n^k + a_1(w) z_n^{k-1} + \cdots + a_k(w)$$

where $a_i \in \mathfrak{m}_0^{(n-1)}$

Theorem 12.3 (Weierstrass preparation theorem) Suppose f is regular of order k in z_n, i.e. $f(0, z_n)$ has a zero of order k at $z_n = 0$ (In general, we can rotate). Then there exists a unique Weierstrass polynomial p such that $f = up$ for a unit u

Proof. By Rouché's theorem, on a small neighborhood of $0 \in \mathbb{C}^n$, $z_n \mapsto f(w, z_n)$ has k-zeros $\xi_1(w), \cdots, \xi_k(w)$, counting multiplicity. By the residue theorem, for each $r = 1, 2, \cdots$ and fixed w

$$\sum_{j=1}^k \xi_j(w)^r = \frac{1}{2\pi i} \int_{|\ \xi\ | = \delta} \zeta^r \frac{f'(w,\zeta)}{f(w,\zeta)} d\zeta$$

From this, the elementary symmetric polynomials in the $\xi_j(w)$ are holomorphic in w. Hence

$$p(w, z_n) = \prod_{j=1}^k (z_n - \xi_j(w))$$

is a Weierstrass polynomial. Now let $u = f/p$. This is holomorphic by the Riemann extension theorem

□

Theorem 12.4 (Weierstrass division theorem) Let $p \in \mathcal{O}_0^{(n-1)}[z_n]$ be a Weierstrass polynomial of degree k. Then any $f \in \mathcal{O}_0^{(n)}$ can be written uniquely as $f = gp + r$, where $r \in \mathcal{O}_0^{(n-1)}[z_n]$ is a polynomial of degree $< k$. If $f \in \mathcal{O}_0^{(n-1)}[z_n]$, then so is g

Proof.

□

Lemma 12.5 A Weierstrass polynomial $h \in \mathcal{O}_0^{n-1}[z_n]$ is reducible over $\mathcal{O}_0^{(n)}$ iff it is reducible over $\mathcal{O}_0^{n-1}[z_n]$. In this case, all factors are Weierstrass polynomials, modulo units

Proof.

□

Theorem 12.6 $\mathcal{O}_0^{(n)}$ is a UFD

Proof.

□

Theorem 12.7 (Oka) $\mathcal{O}_0^{(n)}$ is a Noetherian local ring

Proof. By Hilbert basis theorem, $\mathcal{O}_0^{(n-1)}[z_n]$ is Noetherian by induction. Let $\mathfrak{a} \subseteq \mathcal{O}^{(n)_0}$ be an ideal, and assume $g \in \mathfrak{a}$ is a regular in z_n. Up to units, we may assume it is a Weierstrass polynoimal Let $g_1, \cdots, g_m$ generate $\mathfrak{a} \cap \mathcal{O}_0^{(n-1)}[z_n]$. We claim the $g, g_1, \cdots, g_m$ generate $\mathfrak{a}$. Let $f \in \mathfrak{a}$. By WD write $f =$

□

We will be interested in $\mathcal{O}_0$ modules $\mathcal{E}_0$

Since $\mathcal{O}_0$ is Noetherian, a finitely generated module is finitely presented, i.e. there is an exact sequence (This is actually called *syzygy*)

$$\mathcal{O}_0^{\oplus s} \to \mathcal{O}_0^{\oplus r} \to \mathcal{E}_0 \to 0$$

$\mathcal{O}_0^{\oplus s}$ are the finitely many relations

Let $\Omega \subseteq \mathbb{C}^n$ be a domain, and $F^1, \cdots, F^r \in A(\Omega)^{\oplus p}$. For $\xi \in \Omega$, let $\mathcal{E}_\xi$ be the $\mathcal{O}_\xi$ module generated by $F_\xi^1, \cdots, F_\xi^r$. Let $\mathcal{R}_\xi$ be the $\mathcal{O}_\xi$ module of relations

$$\mathcal{R}_\xi = \{\}$$

Since $\mathcal{O}_\xi$ is Noetherian, $\mathcal{R}_\xi$ is finitely generated submodule of $\mathcal{O}_\xi^{\oplus r}$ for every $\xi \in \Omega$ Question: Can we find finitely many functions

We now rephrase Oka's theorem

Theorem 12.8 Given, then for each there is an open set and such that

$$(\mathcal{O}_U^{(n)})^{\oplus r} \xrightarrow{\lambda} (\mathcal{O}_U^{(n)})^{\oplus p} \xrightarrow{\mu} (\mathcal{O}_U^{(n)})^{\oplus q}$$

is exact on U, here r might depend on the choice of ξ_0

Let $\mathcal{O}_{0,d}^{(n-1)}[z_n]$ denote the polynomials of degree at most d. This is a finitely generated $\mathcal{O}_0^{(n-1)}$ submodule of $\mathcal{O}_0^{(n-1)}[z_n]$

First assume $q = 1$

Lemma 12.9 Suppose λ, μ have entries in $\mathcal{O}_0^{(n-1)}[z_n]$ of degree l and m, respectively. Assume that one of the entries of μ is regular in z_n. Then we have a sequence

$$(\mathcal{O}_{0,d-l}^{(n-1)}[z_n])^{\oplus r} \xrightarrow{\lambda} (\mathcal{O}_{0,d}^{(n-1)}[z_n])^{\oplus p} \xrightarrow{\mu} (\mathcal{O}_{0,d-l}^{(n-1)}[z_n]) \tag{12.1}$$

as well as

(12.2)

If $d \geq m$ and is an exact suquence of modules, then is an exact sequence of modules

Proof. Assume $\mu =$

□

Consequence of Oka's theorem: coherent analytic sheaves on X form an abelian category. This is not true for holomorphic vector bundles

Proposition 12.10 If $\phi : \mathcal{E} \to \mathcal{F}$ is a homomorphism between coherent analytic sheaves, then $\ker \phi$ and coker ϕ are coherent

Proposition 12.11 Suppose we have a short exact sequence of analytic sheaves

$$0 \to \mathcal{S} \to \mathcal{E} \to \mathcal{Q} \to 0$$

If any two of the three sheaves are coherent, then so is the third

13 Cartan theorems

Theorem 13.1 $K \subseteq X$ compact with $\hat{K} = K$. For all coherent sheaves $\mathcal{E}$ defined on a nbd of K

1. There exist sections of $\mathcal{E}$ on a nbd U of K that generate $\mathcal{E}_z$ for every $z \in U$
2. and if σ is any other section of $\mathcal{E}$ on U, then $\sigma = \sum f_i \sigma_i$ for some $f_i \in A(U)$

In terms of exact sequences, there exists on U an exact sequence $O \to E \to 0$ and the induced map is exact

Lemma 13.2 $\hat{K} = K$ as above, $f \in A(X)$. Suppose property holds for $K_a = \{z \in K \mid \text{Re } (f(z)) = a\}$ for all $z \in \mathbb{R}$. Then it holds for K

Proof. Define $K_{a,b} = \{z \in K \mid a \leq \text{Re } (f(z)) \leq b\}$, then $\hat{K}_{a,b} = K_{a,b}$. Indeed, this condition is equivalent to saying $\mid e^{f(z)} \mid \leq e^b$, and $\mid e^{-f(z)} \mid \leq e^{-a}$. Suppose $z \in \hat{K}_{a,b}$. If $z \in K$, then $\mid e^{f(z)} \mid \leq \| e^{f(z)} \|_{K_{a,b}} \leq e^b$, $\mid e^{-f(z)} \mid \leq \| e^{-f(z)} \|_{K_{a,b}} \leq e^{-a}$. If $z \notin K = \hat{K}$, there is $g \in A(X)$ such that $\mid g(z) \mid \geq \| g \|_K \geq \| g \|_{K_{a,b}}$, $z \notin \hat{K}_{a,b}$. Suppose $\beta = \sup$. If $\beta = +\infty$, then we are done. If not, by hypothesis we can find a nbd U_1 of $K_{\beta,\beta}$ and sections $\phi_1, \cdots, \phi_p$ generating $\mathcal{E}$ on U_1. Choose $\alpha < \beta < b$ such that $K_{a,b} \subseteq U_1$. We can find nbd U_2 of $K_{-\infty,a}$ and sections $\psi_1, \cdots, \psi_q$ generating $\mathcal{E}$ on U_2. Regard ϕ, ψ as column vectors, shrinking the sets if necessary, we may assume there are matrix valued holomorphic functions A_1, A_2 such that $\psi = A_1\phi$ and $\phi = A_2\psi$. In other words

$$\begin{bmatrix} \phi \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -A_1 & 1 \end{bmatrix} \begin{bmatrix} 1 & A_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \psi \end{bmatrix}$$

We wnat to find vectors f_1, f_2 of dimension $p + q$ such that

$$f_1 \cdot \begin{bmatrix} \phi \\ 0 \end{bmatrix} = f_2 \cdot \begin{bmatrix} 0 \\ \psi \end{bmatrix}$$

on the intersection $U_1 \cap U_2$. Let

$$c_{21} = c_{12}^{-1} = \begin{bmatrix} 1 & \\ A_2^T & 1 \end{bmatrix} \begin{bmatrix} 1 & -A_1^T \\ & 1 \end{bmatrix}$$

Then f_i satisfy $f_2 = c_{21} f_1$. We may view $\{f_1, f_2\}$ as a section of rank $p + q$ holomorphic vector bundle over $U_1 \cap U_2$ with transition function c_{12}. We may also suppose that $U_1 \cap U_2$ is Stein. Since we have already show that global sections of vector bundles exist, we can find such a pair and this leads to a contradiction

□

Theorem 13.3 (Cartan Theorem A) Let $\mathcal{E} \to X$ be a coherent analytic sheaf on a Stein manifold. Then for every $z \in X$, $\mathcal{E}_z$ is generated by the germs of global sections $\Gamma(X, \mathcal{E})$

Theorem 13.4 (Cartan Theorem B) Let $\mathcal{E} \to X$ be a coherent analytic sheaf on a Stein manifold. Then $H^q(X, \mathcal{E}) = \{0\}$ for all $q \geq 1$

Remark 13.5 For locally free sheaves, $H^q(X, \mathcal{E}) \cong H^{0,q}_{\bar{\partial}}(X, E)$, and we have proven this result using the solution to the $\bar{\partial}$ equation

Definition 13.6 $\mathcal{E} \to X$ is called a *fine sheaf* if it admits a partition of unity for every locally finite cover

Proposition 13.7 Suppose

$$0 \to \Gamma(X, \mathcal{S}) \to \Gamma(X, \mathcal{E}) \to \Gamma(X, \mathcal{Q})$$

is exact. If $\mathcal{S}$ is a fine sheaf, then is $\Gamma(X, \mathcal{E}) \to \Gamma(X, \mathcal{Q})$ surjective

Definition 13.8 Cohomology is a funtor

$$H^\bullet : \{\text{sheaves on } \ X\} \to \{\text{abelian groups}\}$$

satisfying axioms

1. $H^0(X, \mathcal{E}) = \Gamma(X, \mathcal{E})$
2. If $\mathcal{E}$ is a fine sheaf, then $H^p(X, \mathcal{E}) = 0$ for $p \geq 1$
3. If $0 \to \mathcal{S} \to \mathcal{E} \to \mathcal{Q} \to 0$ is a short exact sequence, then we have long exact sequence

Proposition 13.9 Let $\mathcal{E} \to X$ be a sheaf with a fine resolution, i.e. a long exact sequence

$$0 \to \mathcal{E} \to \mathcal{A}^0 \overset{d_0}{\to} \mathcal{A}^1 \overset{d_1}{\to} \cdots$$

Then

$$H^p(X, \mathcal{E}) = \frac{\ker(\mathcal{A}^p(X) \overset{d_p}{\to} \mathcal{A}^{p+1}(X))}{\text{im } (\mathcal{A}^{p-1}(X) \overset{d_{p-1}}{\to} \mathcal{A}^p(X))}$$

Theorem 13.10 Let X be a complex manifold, and suppose that $H^1(X, \mathcal{E}) = 0$ for every coherent analytic sheaf $\mathcal{E} \to X$. Then X is Stein

Proof. $A(X)$ separates points: Let $\mathcal{I}_{x,y}$ be the sheaf of germs of holomorphic functions vanishing at x and y. This is a coherent sheaf. From the exact sequence

$$0 \to \mathcal{I}_{x,y} \to \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}_{x,y} \to 0$$

We have by the assumption

$$H^0(X, \mathcal{O}_X) \to H^0(X, \mathcal{O}_X/\mathcal{I}_{x,y}) \cong \mathcal{C}^{\oplus 2} \to H^1(X, \mathcal{I}_{x,y}) = 0$$

Hence there is $f \in A(X)$ such that $f(x) = 1$ and $f(y) = 0$
$A(X)$ gives local coordinates: Let $z_1, \cdots, z_n$ be local coordinates at x, then the germs of these generate $\mathcal{I}_x/\mathcal{I}_x^2$. we have exact sequence

$$0 \to \mathcal{I}_x^2 \to \mathcal{I}_x \to \mathcal{I}_x/\mathcal{I}_x^2 \to 0$$

From the vanishing of $H^1(X, \mathcal{I}_x^2)$, we may find functions $f_1, \cdots, f_n \in A(X)$, vanishing at x, $f_i = z_i$ modulo
X is holomorphically convex: $\{x_i\}$ is a discrete sequence, $\mathcal{I}_{\{x_i\}}$ is coherent. From the vanishing of $H^1(X, \mathcal{O}_X/\mathcal{I}_{\{x_i\}})$, there is $f \in A(X)$ such that $f(x_i) = i$

□

Remark 13.11 Notice the similarity with the proof of the Kodaira embedding theorem

Theorem 13.12 (Cartan-Serre theorem) X is a compact complex manifold. Then for any coherent sheaf $\mathcal{E} \to X$, the cohomology groups $H^p(X, \mathcal{E})$ are finite dimensional

Proof. Recall the Frechet space structure we have on sections of coherent sheaves. Coherent sheaves satisfy the Montel property: if $U \subset\subset V$, then the restriction map $H^0(V, \mathcal{E}) \to H^0(U, \mathcal{E})$ is a compact operator. Now choose finite Stein open coverings $\{U_i\}, \{V_i\}$, $V_i \subset\subset V_i$ such that

$$H^p(\mathcal{U}, \mathcal{E}) = H^p(\mathcal{V}, \mathcal{E}) = H^p(X, \mathcal{E})$$

Then by the finiteness of the cover, the restriction map

$$\rho : C^p(\mathcal{V}.\mathcal{E}) \to C^p(\mathcal{U}, \mathcal{E})$$

is compact. The coboundary map δ is clearly continuous, so $Z^p = \ker \delta$ is a closed subspace, hence a Frechet space. Now define

$$u : Z^p(\mathcal{U}, \mathcal{E}) \oplus C^{p-1}(\mathcal{U}, \mathcal{E}) \to Z^p(\mathcal{U}, \mathcal{E})$$

by $u(\phi, \psi) = \rho(\phi) + \delta\psi$. Then u is surjective. Apply Theorem 13.13 to $T = u$ and $S = -\rho$

□

Theorem 13.13 (Laurent Schwartz) Let E, F be Frechet spaces with $T : E \to F$ is surjective and $S : E \to F$ compact. Then $T + S$ has closed range and $F/\text{ im }(T + S)$ is finite dimensional

Definition 13.14 X is a complex manifold. A subset $V \subseteq X$ is called an *analytic subvariety* of X is for every point $x \in X$ there is neighborhood U of x and $f_1, \cdots, f_l \in A(U)$ such that $V \cap U = Z(f_1, \cdots, f_l)$, the common zero locus of f_i's

It follows that $V \subseteq X$ is closed. A complex submanifold of codimension l is an analytic subvariety

The ideal sheaf $\mathcal{I}_V$ is coherent by Oka's theorem. $\mathcal{O}_V = \mathcal{O}_X/\mathcal{I}_V$

Fact: The maximum principle holds for holomorphic functions on subvarieties

Consequence: The only compact subvarieties of a Stein manifold are finite collections of points

References

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